

Towards worker-centric construction scene understanding: Status quo and future directions

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ABSTRACT

Construction scene understanding is the process of perceiving, analyzing, and interpreting three-dimensional dynamic scenes observed through sensor networks, which is usually real-time. The purpose is to understand the construction scene by analyzing the geometric and semantic features of the objects and their relationships. Construction scene understanding is a basic technology for construction automation and Human-Robot Interaction. Although there are several reviews on this topic, existing reviews do not cover the latest technological advances and lack systematic elaboration for construction scene understanding. Therefore, this paper reviews research on construction scene understanding conducted over the past fifteen years, summarizing five key scene elements: signals, pixels, objects, relationships, and events, along with the richness of semantic information. It also outlines advancements in perception and reasoning abilities in this domain. In addition, this review proposes five research trends and seven future directions to provide some inspiration for researchers.

1. Introduction

Scene understanding plays a pivotal role in the construction industry, supporting vision-based automation tasks and facilitating information management workflows, such as safety monitoring and facilities management. The goal of scene understanding is to help computers fully understand real-world scenes as humans do. In recent years, the urgent need to address practical problems in construction and the increasing maturity of deep learning have driven the rapid development of computer vision in this field. Artificial Intelligence (AI), especially deep learning, has become the dominant approach in computer vision and is increasingly applied in construction to process visual data, with the use of photos and videos on construction sites growing steadily [1]. The rapid development of computer vision and abundant visual data of construction provide opportunities for developing construction scene understanding, which helps to improve construction safety [2–7], productivity [8–12], progress [13,14], quality [15], and resource management [16–18]. The rise of Large Language Models (LLM) also brings new opportunities for scene understanding, promoting the fusion of language and vision, and facilitating tasks such as Image Captioning (IC), Visual Question Answering (VQA), visual reasoning, cross-modal learning.

In construction, the scene refers to a specific situation, place, and series of actions that workers complete tasks, involving objects such as materials and machinery, as well as background such as space and environment. Workers are the core of the construction scene, with subjective initiative, constantly moving and evolving. They are important in construction safety, progress, and quality management. Most importantly, workers are constantly interacting with the ever-changing construction scene and producing dynamic results. Moreover, worker safety is crucial to ensuring overall construction safety. Therefore, worker-centric scene understanding is a key challenge in this field. Addressing this issue will advance construction robots and scene understanding focused on machines and materials.

Due to the small and numerous objects and complex relationships on construction sites, construction scene understanding is a complex problem and researchers have spent a lot of effort on related research. There are several reviews on construction scene understanding [1,11,19–21]. From a technical perspective for scene understanding, in 2021, Arashpour et al. [1] conducted a review based on underlying features (such as color, texture, and shape) for classical image processing. In 2022, Dang et al. [19] reviewed AI-based sound perception methods, highlighting that research on auditory situational awareness is still in progress. Rao et al. [11] reviewed sensors, methods, and

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0926-5805/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

applications for construction site monitoring, focusing on technologies for real-time mapping, scene understanding, positioning, and tracking construction activities in both indoor and outdoor environments. In 2023, Mahmoudi et al. [20] reviewed and compared deep learning models related to Action Recognition (AR), focusing on accuracy, interpretability, computation time, memory size, and flexibility. Zhou et al. [21] comprehensively reviewed construction safety management from three perspectives: dataset, machine learning approach, and research topic. However, existing reviews do not cover the latest technological advances. There is no universally accepted definition of scene understanding, nor a systematic elaboration of the concept. New technological advances provide new possibilities for advancing construction scene understanding. With the development of computer vision, the techniques used for scene understanding have gradually shifted from classical image processing [22] to deep learning [23]. For scene understanding, most studies are limited to a specific subtask such as object detection [22], semantic segmentation [18], depth estimation [23], or relationship detection [17], which only involves the perception ability of scene understanding. Only a few studies have addressed the reasoning ability of scene understanding [4]. Besides, most current reviews on construction scene understanding focus on a single subtask.

In recent years, Large Language Models (LLM) and generative AI have flourished. The rapid iteration of related technologies has impacted traditional research paradigms such as computer vision and natural language processing. In this context, research on construction scene understanding from the perspective of workers remains valuable in the following two aspects. First, technical difficulties such as multi-view fusion, small object detection, and Situation Awareness (SA) constrain the development of computer vision. Although generic Large Language Models (LLM) can solve some of these problems, the embedding of industry-specific knowledge plays a crucial role in overcoming these technical difficulties and improving the accuracy and stability of the models. Secondly, this research can provide a new research paradigm for understanding the causal chain of construction accidents, offering a fresh perspective for comprehending worker behavior, assessing worker safety knowledge, and automatically detecting the evolving process of accident occurrence.

Due to the development of scene understanding and large language models in recent years, it is necessary to conduct a new review on this topic. The paper provides a comprehensive review of worker-centric construction scene understanding. It identifies research hotspots and challenges, analyzes development trends, and addresses gaps in current research, offering new insights into understanding worker behavior and accident causation. The paper is organized as follows, as illustrated in Fig. 1: Section 2 presents the research methodology and an overview of construction scene understanding, while Section 3 outlines research

development of construction scene understanding, detailing perception ability and reasoning ability. In Section 4, this study analyzes five major trends in construction scene understanding. Finally, in Section 5, current research challenges are discussed and future research directions are given.

2. Research approach

2.1. Material and methods

This study employs a content analysis-based review method, which is a well-established approach widely used in engineering and construction management research. Relevant articles in the target area were identified through Web of Science and Scopus. Topics, research fields, and journals were screened to ensure the relevance and quality of the selected research literature. Specifically, there are three stages, as shown below (Fig. 2):

- (1) An exhaustive search was conducted using both the Web of Science and Scopus databases, covering the period from 2005 to 2024. To ensure comprehensive access to relevant literature, various search keywords were applied: scene understanding, relationship detection, event detection, image caption, visual semantics, scene perception, situation awareness, situational awareness, construction, and worker. The search focused on titles, abstracts, and keywords to identify and extract relevant articles. To refine the scope of the results, a screening process was applied based on two criteria: 1) documents written in English, and 2) studies included in the Web of Science Core Collection (WOSCC) or Scopus. A total of 218 documents were extracted, including journal papers and conference papers.
- (2) In the second stage, documents that did not include the above keywords in the title or abstract were manually screened. This phase focused on a thorough review of abstracts to identify articles specifically related to construction scene understanding, resulting in the final selection of 169 articles for further analysis.
- (3) In the third stage, after reviewing the article content, irrelevant papers were filtered out, leaving a total of 125 publications for further analysis. The two criteria for content review during this phase were: 1) a focus on worker-centric construction scene understanding, and 2) an emphasis on relationships and events in construction, with attention to high-level semantics.

While the search method may not guarantee comprehensive coverage of all relevant papers, it is sufficient to provide a substantial number of significant state-of-the-art works, from which the review can

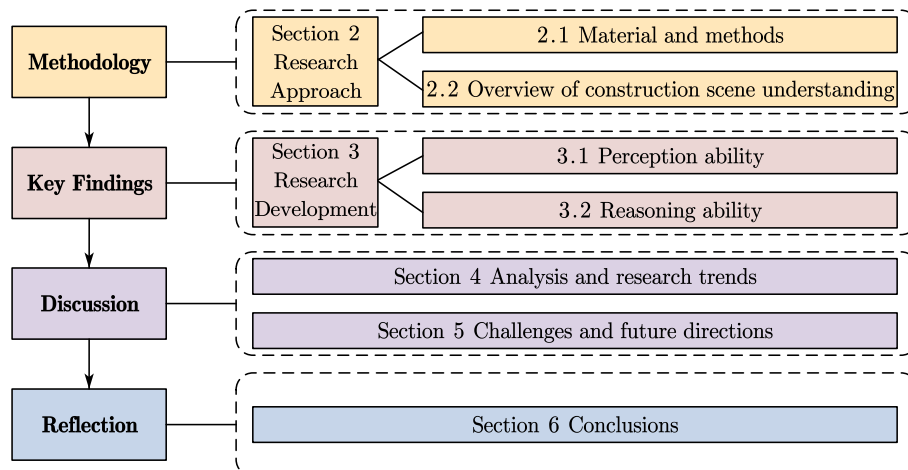


Fig. 1. Structure of the paper.

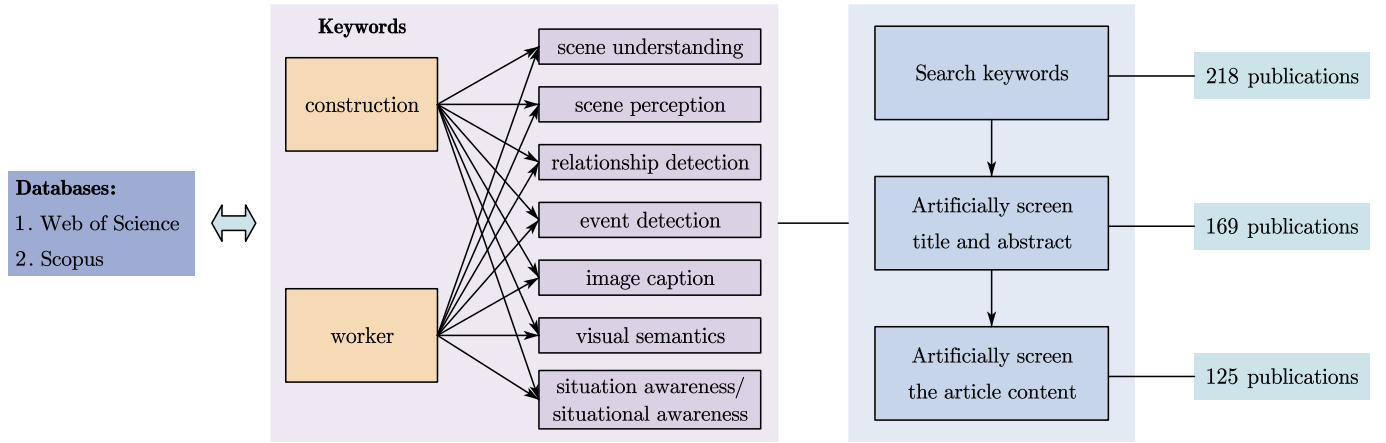


Fig. 2. Framework for content analysis-based review.

draw general conclusions and recommend future research.

2.2. Overview of construction scene understanding

Construction scene understanding refers to the real-time process of perceiving, analyzing, and interpreting three-dimensional dynamic scenes captured by sensor networks. The data sources typically include images, text, point clouds, and others. Data processing often involves steps such as feature extraction, relationship identification, semantic feature integration, and multimodal data fusion. The ultimate goal is to gain a comprehensive understanding of the construction scene by analyzing both the geometric and semantic features of objects and their relationships, thereby supporting engineers in addressing practical construction tasks. A complete and mature scene understanding framework needs to have two basic abilities [24,25]: (1) Perception abilities: the ability to automatically extract key information (such as objects, attributes, relationships, and events) from multimodal signals (such as visible or infrared images, videos, point clouds, sounds, radar, GPS) for scene understanding. (2) Reasoning abilities: based on the extracted scene information, combine it with expertise specific to the construction scene to guide the decision-making of the autonomous system. Scene understanding frameworks deliver valuable information

to autonomous systems quickly, accurately, and systematically, without human intervention, supporting the automation of systems such as construction robots [26].

Scenes can involve many different types of physical objects (such as workers, materials, and equipment) interacting with each other or their environment. They may last only a few moments (such as a person falling) or stretch over months (such as the construction of a building), and they can range from a single wall to an area that spans an entire city. Sensors include not only cameras (such as omnidirectional cameras and infrared cameras) but also microphones and other types of sensors (such as RGB-D, LIDAR, radar, IMU, and GPS). Integrating multiple types of sensors is usually more effective than relying on single-type sensors, as it helps overcome the limitations of single-type sensors. Scene understanding combines three key domains: computer vision, cognition, and software engineering. It is widely applied in areas such as autonomous driving, assistive robotics, intelligent transport, smart cities, entertainment, and healthcare. However, compared to other industries, construction scenes have unique characteristics (such as unknown objects, dynamic objects, background clutter, occlusion, pose variations, and scale variations), which can severely hinder the efficiency and accuracy of scene understanding.

Elements in scene understanding are represented through five levels:

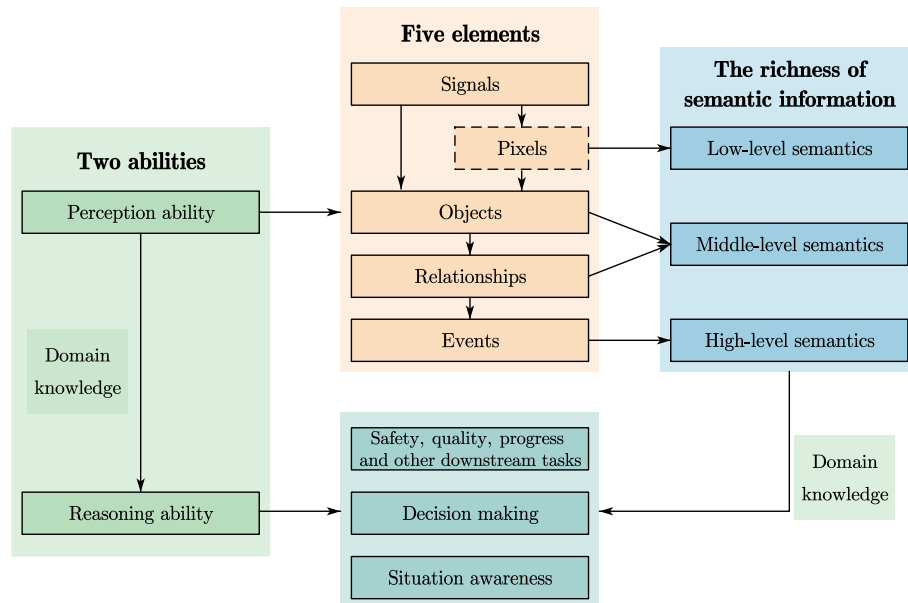


Fig. 3. Framework for construction scene understanding.

signals, pixels, objects, relationships, and events. The five elements are related through a process of representation, as illustrated in Fig. 3. Signals can be derived from sensors and combinations of sensors. After data processing, the signals can be represented as interpreted elements—namely objects, relationships, and events—that can be understood by humans. Specifically, for visual signals, pixels serve as the basic units that form an image. The aggregation of pixels forms visual objects, relationships, and events in the image. Additionally, pixels carry a limited amount of interpreted semantic information, such as RGB values that represent color. Within the context of semantics, objects represent distinct entities in a scene, relationships describe the interactions between these objects, and events capture the dynamics or changes occurring within the scene. These three elements are interconnected and form a semantic hierarchy: objects serve as the fundamental building blocks, while relationships and events provide contextual meaning to the interactions between objects. Prior studies, such as those by Clowes [27], have offered perspectives on the semantics of pictures. “The interpretation of pictures involves a further syntactic structure of the event being pictorially depicted which constitutes the semantics of the picture [...] we should regard the semantics of pictures as concerned with the exhibition of relationships of a non-pictorial kind. These relationships must be regarded as defining the character of the ‘physical world’. Pictures express events in this physical world.” [27] Informed by advancements in technology and a synthesis of relevant literature, this study categorizes semantics into three levels according to the richness of semantic information: low-level semantics, which pertains to basic features (such as color or shape); middle-level semantics, which involves more complex information, such as objects and their relationships; and high-level semantics, which encompasses the interpretation of events within a scene. This structure facilitates the progressive representation of raw signals into more meaningful concepts.

The low-level semantics focus on pixel-level features, usually general features such as color, texture, and shape. In computer vision tasks relying on visible light, it is indeed true that both color and shape features at the pixel level are affected by lighting conditions. For instance, insufficient illumination can increase image noise, reduce contrast, and diminish color fidelity, impairing the reliability of these features. Conversely, excessive illumination may cause overexposure, loss of detail, and color saturation artifacts, further complicating their detection and representation. However, transformations such as rotation, scaling, and perspective changes are common in data acquisition processes due to variations in camera angles and object orientations. For 2D images, color features are more stable and robust than shape features under such transformations. Assuming consistent lighting and camera parameters, an object's color remains unchanged during such transformations, unlike shape-related features. Due to its advantages, color is used as a key feature in vision-based applications in construction. Brilakis et al. [28] accomplished work in image processing by using filters to decompose an image into several types of basic features (such as color and texture). The pixels of each image are grouped into meaningful clusters and then matched with pre-classified material samples. Through these operations, the detection, retrieval, and indexing of materials within an image can be achieved. The information obtained from low-level feature extraction is generalized. While basic image retrieval and classification can be achieved using these features, relying solely on them reduces the accuracy of retrieval and classification and limits complex scene understanding, such as the unsafe behavior of workers. Middle-level semantics focuses on objects and relationships, including tasks such as image classification, object detection, semantic segmentation, and relationship detection. However, the limited information is insufficient for autonomous systems, which require a more comprehensive understanding of their surroundings [29]. High-level semantics focuses on complex information such as events, environments, and states. High-level semantic extraction incorporates methods like pattern recognition to establish semantic associations between different objects. This enables the completion of specific tasks such as event recognition,

behavior recognition, and state perception. With the advancement of deep learning, convolutional neural networks can effectively extract high-level semantics from images and are widely used in construction to identify unsafe behaviors [30]. Feature extraction allows for the perception of objects, relationships, and events. However, these alone are insufficient to enable reasoning ability for scene understanding. Scene semantics, such as objects, relationships, and events, need to be combined with domain-specific knowledge [31] to further achieve scene understanding and support decision-making in construction.

In recent years, construction scene understanding has been a hot topic for practitioners and researchers. Scene perception units have gradually developed from pixels to objects, relationships, events, and scenes. Perception methods have gradually developed from classical image processing and machine learning to deep learning and end-to-end learning, and scene understanding has gradually developed from perception ability to reasoning ability. However, the existing perception methods have many defects (such as insufficient interpretability and difficulty in applying rules), which cannot meet the increasingly complex engineering needs. This is because existing methods do not fully account for the interdependence between scenes and objects, or between objects within the construction environment. There are three problems: (1) The scene-object coupling mechanism is unclear, the relationships between objects are not well understood, and there is a lack of high-level feature extraction, making it difficult to model semantics and accurately perceive construction scenes. (2) Knowledge modeling is inadequate, making it challenging to embed rule constraints and perform knowledge reasoning during the perception process, which leads to poor interpretability of the perception ability. (3) The self-learning mechanism of the perception algorithm is unclear, and it is not well integrated with application requirements. As a result, the algorithm has low reusability, weak adaptability, and poor scene flexibility, making it difficult to support perception in complex and varied construction environments. The above issues have become bottlenecks that limit construction scene perception. They are key challenges that must be addressed to improve construction scene understanding.

2.3. Contents of the review

Given the characteristics of construction scenes, the contents of this review is shown in Fig. 4. The focus is on the two abilities of scene understanding, with further discussion on research trends and future directions.

3. Research development of scene understanding

Scene understanding includes a series of tasks on perception ability, which can be classified according to different elements, such as object detection, semantic segmentation, object tracking, depth estimation, gesture estimation, relationship detection, and event detection. According to the needs and characteristics of construction scenes, construction scene understanding can be broadly classified into perception ability and reasoning ability at the application level, as outlined in Fig. 5. Perception ability includes entity-based scene understanding, relationship-based scene understanding, and event-based scene understanding. Firstly, entity-based scene understanding needs to identify specific entities in static or dynamic construction scenes, such as workers, helmets, PPEs, and trucks. Various advanced object detection algorithms are widely used to identify entities in construction sites. In addition, object-tracking algorithms are used to detect the positions and dynamic trajectories of various entities in videos. Second, relationship-based scene understanding builds on entity-based scene understanding, focusing on the semantic representation of relationships between workers, materials, and machinery on construction sites. Thirdly, event-based scene understanding focuses mainly on the recognition of workers' behaviors or events in construction sites. Reasoning ability relies on perception ability and combines specific expertise [31] to

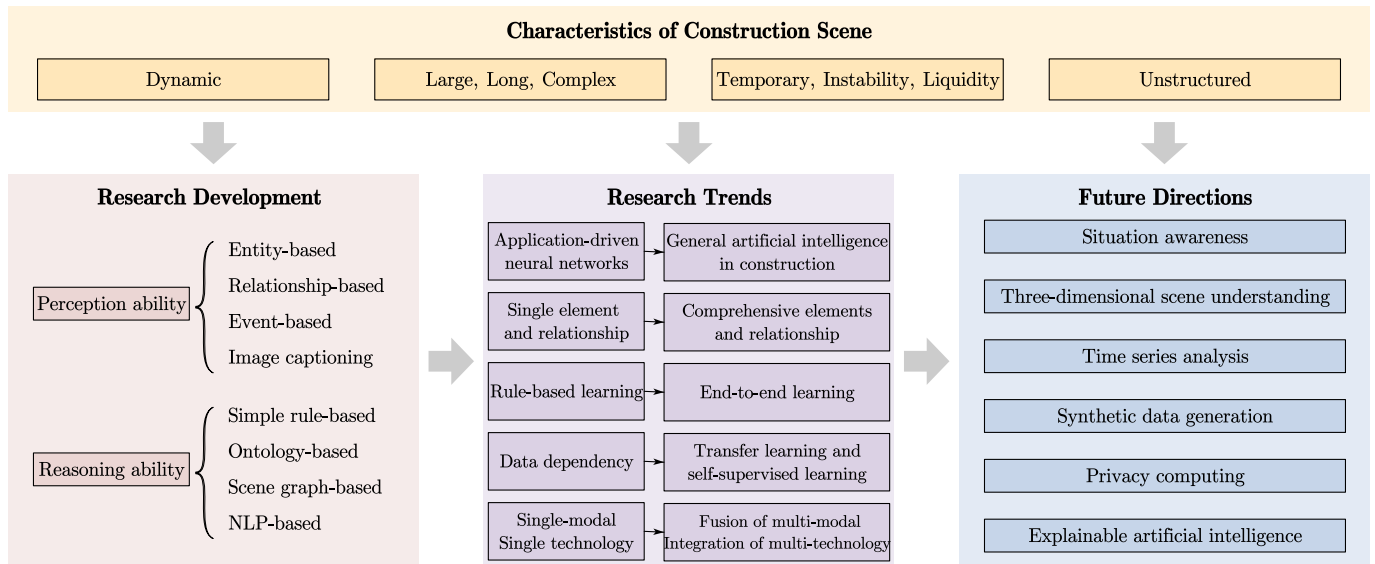


Fig. 4. Contents of the review.

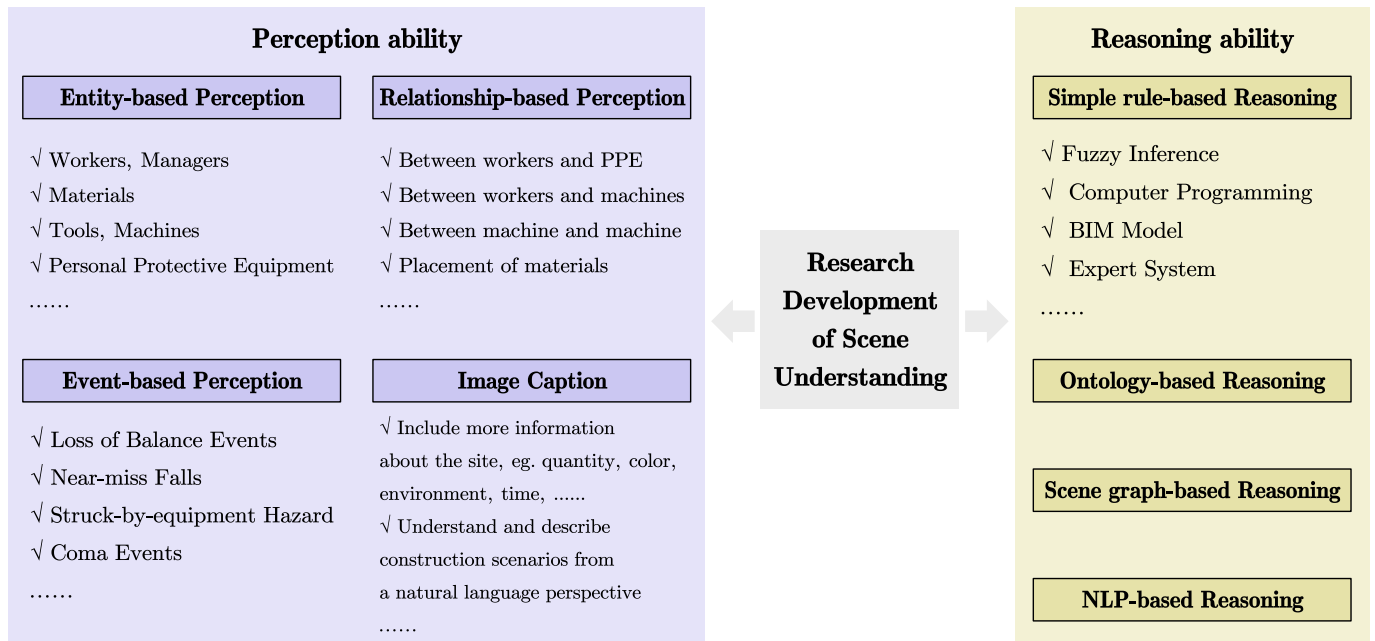


Fig. 5. Research development in construction scene understanding: from perception ability to reasoning ability.

analyze and make judgments about the perceived information, ultimately guiding the decision-making of autonomous systems in construction. Fig. 6 presents the progression of key research methods in construction scene understanding.

3.1. Perception ability

3.1.1. Entity-based perception

Entity-based perception is the earliest research focus and serves as the foundation for more complex perception tasks that follow. As early as 2008, Doerr et al. [32] proposed a 3D feature tracking method for detecting and tracking object features during seismic events, addressing the limitations of 2D tracking approaches. In early research, deep learning has not yet received much attention from researchers, and image processing remains the primary technical tool for related studies. In 2016, Kim et al. [22] developed a method to identify various objects

on construction sites by combining Scale Invariant Feature Transform (SIFT) with Nearest Neighbors (NN). This approach retrieves the most relevant images from the database and then transfers the labels of these images to the query image. Research at this stage has begun to integrate machine learning with image processing and gradually expand applications in machine learning. In 2017, Lundeen et al. [33] used Clustering and Iterative Closest Point (CICP) and Generalized Resolution Correlative Scan Matching (GRCSM) to perceive construction objects, so that construction robots could adapt to unexpected situations and perform high-quality work. Based on hybrid kinematic shape and key node features, Yuan et al. [34] adopted stereo cameras and stereo triangulation to achieve 3D tracking of excavators. With the increasing maturity of deep learning, scene understanding based on deep learning has gained more and more attention. In 2018, Asadi et al. [29] proposed a system that can detect both static and dynamic objects, but better strategies are needed to deal with dynamic objects to improve the

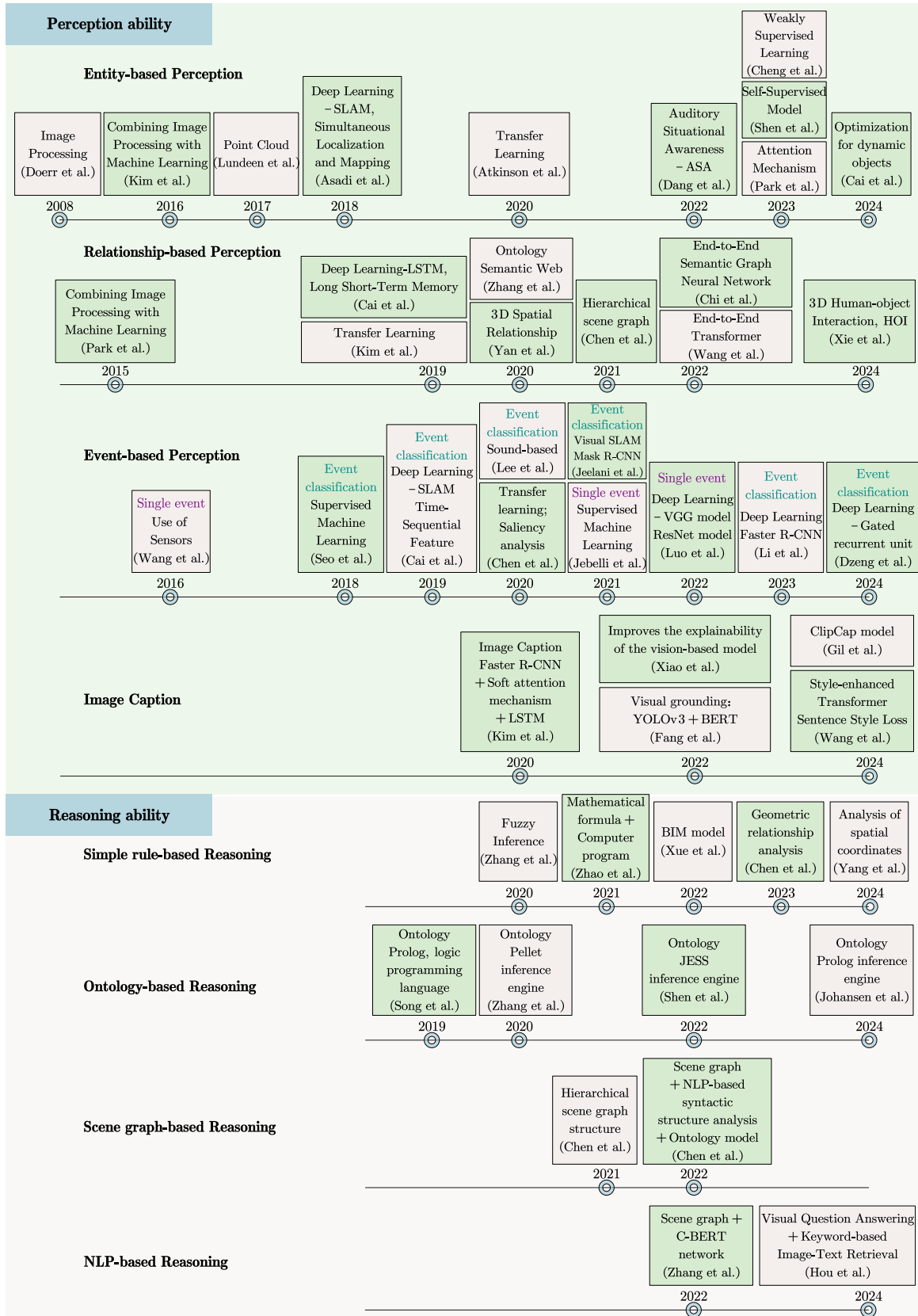


Fig. 6. Progression of key research methods in construction scene understanding.

accuracy and robustness of SLAM. In 2020, Atkinson et al. [35] used Mask R-CNN with Transfer Learning for the segmentation of underfloor scenes. In 2021, Han et al. [36] developed a mobile robotic system using LNS-Net for the pixel-level semantic segmentation of objects. To evaluate the productivity of rebar tying, Zhao et al. [10] developed a detector based on CenterNet to detect five types of objects (worker, Lguardrail, Rguardrail, beam, and column) and their coordinates in real time. Ning et al. [37] segmented wall contours from point clouds and then reconstructed the wall elevation with detailed openings (door and window). 2D scene understanding has limitations in analyzing spatial relationships. Developing algorithms that handle 3D data and spatial relationships can significantly improve the robustness and accuracy of construction scene understanding. In 2022, Mascaro et al. [38] used Mask R-CNN for 2D semantic segmentation of RGB images and instance-level depth segmentation of depth maps. These results were then matched with objects detected in the RGB images, enabling 3D semantic segmentation. Xue et al. [13] utilize Learning Deeply Supervised Object Detectors from Scratch (DSOD) to identify unfinished building components from the top view. To solve the problem of manual monitoring of helmet-wearing, Yan [39] proposed a helmet-wearing detection algorithm based on face localization. To achieve visual contextual awareness, Zamani et al. [18] adopted Deeplabv3+ to segment the scene containing soil. Most research focuses on visual data and point clouds while ignoring the beneficial effects of other modal data such as auditory. In 2023, Alhebrawi et al. [15] used Random Forest (RF), Gradient Boosted Trees (GBT), and Support Vector Machine (SVM) to analyze the sound waves of an impact hammer, thus identifying crack characteristics such as width, depth, and location. Chen et al. [25] proposed a hybrid visual information analysis framework that trained three types of instance segmentation models (Mask R-CNN, Cascade Mask R-CNN, and YOLACT) to detect the boundary between the worker and the handrail. They also adopted Lightweight OpenPose to estimate the worker's pose. To avoid collision accidents, Lee et al. [40] detected workers and moving equipment (such as carts, trucks, and forklifts) both indoors and outdoors based on YOLOv4. Zhao et al. [12] trained a Faster R-CNN model to identify workers and four high-frequency activities (transfer, process, straighten, and check) in steel reinforcement processing. Among these, recognition of workers is entity-based scene understanding, while recognition of the four high-frequency activities is event-based scene understanding. Park et al. [41] reconstructed the single-stage object detector YOLOv5 using attention mechanism to enhance the performance of detecting workers, ladders, and PPE. They estimated the workers' body poses based on YOLO-Pose and determined whether helmets were being worn correctly based on human keypoints. Seong et al. [42] developed an object detection model based on YOLOv5 for detecting workers, excavators, trucks, mobile cranes, and bulldozers. Shen et al. [23] proposed a priori information-based method that does not require additional sensors. This method recovers the unknown scale and predicts the absolute depth of each pixel in monocular vision. Tran et al. [43] used yolov8 to detect people entering hazardous areas on construction sites. Yin et al. [44] proposed a large-scale point cloud segmentation method using weakly supervised learning to tackle the challenge of labor-intensive labeling. They employed a self-training strategy to generate high-quality pseudo-labels, which expanded the training set and improved segmentation performance. You et al. [45] proposed a novel data enhancement and voting method for stacked object detection using only low-cost sparse sensors. This approach addresses the issue of ambiguous geometric features extracted from raw scan data in stacked object recognition. Zhao et al. [46] used a convolutional neural network to detect workers' locations, thus determining if a worker is entering a hazardous area and providing an early warning. WÓJCIK et al. [47] used a Generalized Region-based Convolutional Neural Network to detect workers and helmets. Xiong et al. [48] introduces a lifelong learning framework for detecting and understanding construction resources. They leverage a unified dataset (OpenConstruction), a common taxonomy, and an informativeness-

based learning algorithm to enable continuous model adaptation without catastrophic forgetting. In 2024, Cai et al. [49] proposed the LSTM-MTL model for human intention prediction, and the LSTM-MTL model takes the worker's current pose and the manipulated object as inputs. This means that accurate perception of the current scene is required before predicting intentions, and the accuracy of scene understanding will directly impact the accuracy of the subsequent intention prediction. Cai et al. [26] proposed an unsupervised method for segmenting dynamic objects, to improve the accuracy of real-time localization by reducing the influence of the scene's dynamics. Khan et al. [50] focuses on multi-worker tracking using a framework that integrates pose estimation, ReID models, and data association techniques. Kumar et al. [51] focuses on workers and helmets using Faster R-CNN and data collected by UAVs. Liu et al. [52] focuses on real-time worker detection in complex construction scenes, employing a head-integrated detecting method that adds a head-detecting branch to the Keypoint R-CNN model. Ma et al. [53] focuses on reflective clothing using an improved YOLOv7-tiny algorithm that incorporates omnidimensional dynamic convolution and partial convolution. Soleymani et al. [54] developed a lightweight approach for small tools detection (LSTD). They adopted a deep neural network with Dynamic Feature Extraction (DFE), Integrated Feature Fusion (IFF), and Attention Mechanism to enhance the detection of small construction tools in real time. Xiang et al. [55] focused on detecting safety helmets using a multiscale Swin Transformer architecture. Kim et al. [56] developed an improved YOLOv8 model for real-time detection of personal protective equipment (PPE) on edge devices. The summary of research on entity-based perception is shown in Fig. 7.

The construction scene has a long time span, during which many changes occur. It involves various objects, such as workers, materials, and machines, with complex interactions among them. Worker-centric construction scene understanding pays more attention to downstream tasks such as construction safety [25,39–43,46,57] and work efficiency [10,12]. Therefore, most entity-based perception in the construction scene focuses on workers, personal protective equipment, and heavy machinery. In terms of modal, most research on construction scene understanding focuses on images and point clouds and a few studies have addressed auditory data.

3.1.2. Relationship-based perception

The relationships between entities are crucial for representing and reasoning in scene understanding. Interpreting the interactions and relationships between workers, materials, machinery, and structures on construction sites is vital for complex construction scene understanding. Scene understanding requires not only recognizing entities and capturing their locations but also describing the semantic interactions between them.

In 2015, Park et al. [58] developed a helmet recognition model by using background subtraction to identify workers, extracting HOG features of the helmet, and feeding these features into an SVM. They then defined a candidate region for the helmet and analyzed the relationship between workers and helmets based on a predefined template. In 2019, Cai et al. [8] used positional and attentional cues to identify complex interactions and entities in videos containing various activities. They developed long short-term memory networks (LSTM) to determine whether two entities belong to the same workgroup and to identify the activities in which the workgroup is involved. The method adopts LSTM to learn positional and attentional cues in interactions among multiple objects and to model temporal dependencies between features, instead of relying on predefined, hand-crafted rules and patterns. Compared with traditional hand-crafted rules and templates, this approach better captures the complex and dynamic interaction patterns of construction sites. Fang et al. [59] used a Mask Region-Based Convolutional Neural Network (R-CNN) to automatically detect three types of entities (person, steel support, and concrete support) and further identified the relationship between person and concrete/steel supports, namely unsafe

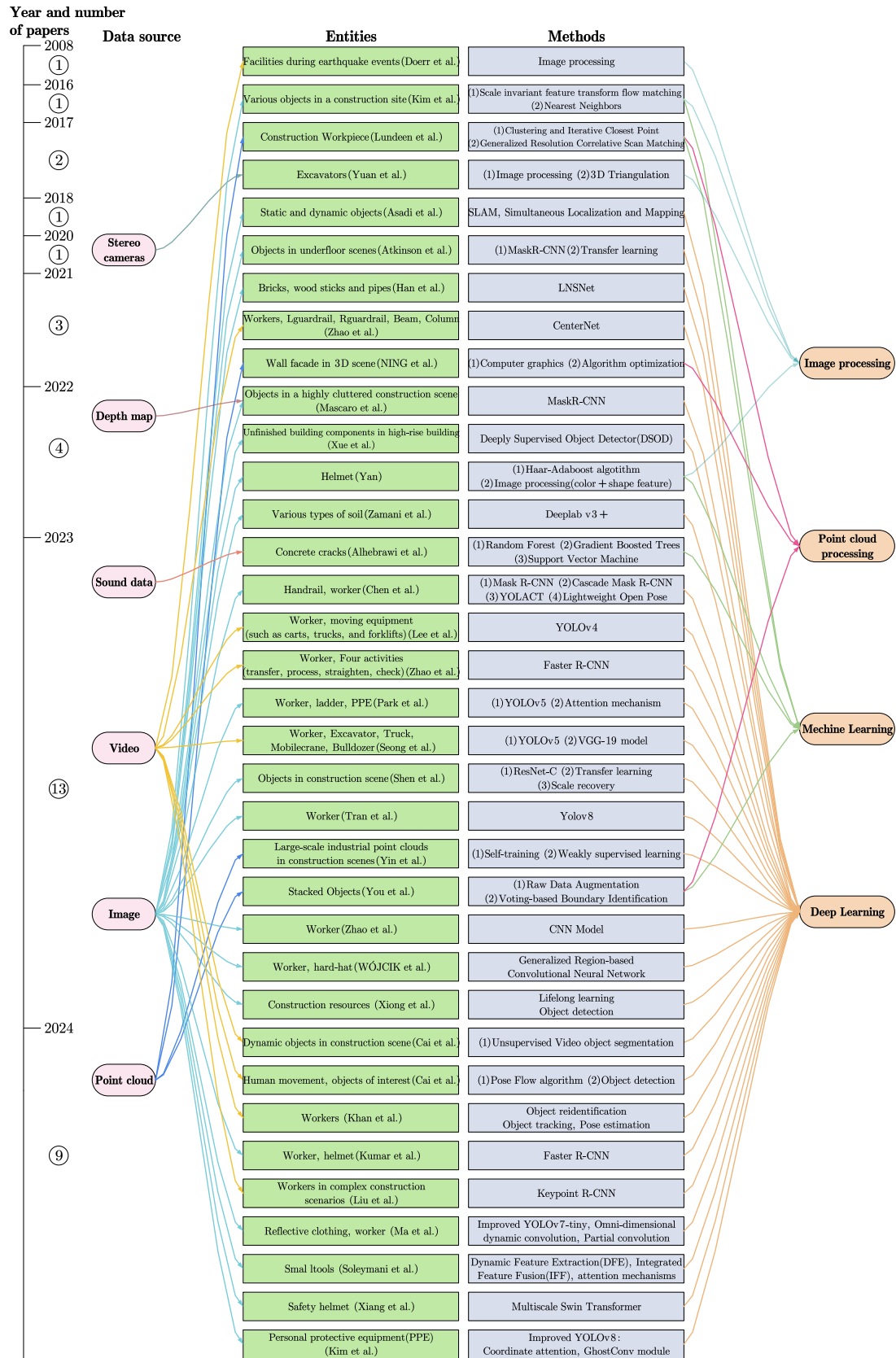


Fig. 7. Summary of research on entity-based perception.

behavior recognition. Kim et al. [60] constructed a unique DNN architecture that can detect entities and their locations as well as describe the semantic relationships between entities. Xiong et al. [24] developed a visual relationship detection module using deep learning to capture entities and their relationships. It represents these relationships as visual triplets in the form (a, b, c), where a and c are the categories of object1 and object2, and b is the relationship predicate. In 2020, Yan et al. [6] reconstructed 3D bounding boxes of objects using video captured from a monocular 2D camera. Then, they estimated the 3D spatial relationship between workers and heavy vehicles and evaluated the proximity and congestion in the construction scene accordingly. Zhang et al. [61] developed a visual extraction module based on R-CNN to identify workers and equipment. Then, they extract centroid coordinates from the detection box, and calculated proximity and crowdedness to represent the spatial relationship between mobile equipment and workers. Based on entity recognition, Zhang et al. [62] further analyzed scene semantics, including object semantics, logical attribute semantics, and spatial semantics. In 2021, Chen et al. [3,4] proposed a scene perception module for detecting improper use of personal protective equipment. They used YOLOv3 to detect various entities (such as hardhats, dust masks, safety glasses, safety belts, and tools) and used OpenPose to extract the joint positions of workers to represent the workers' posture. Geometric relationship analysis was used to achieve individual-object association, triples were used to represent the relationship between workers and PPE, and structured scene graphs were built based on perceived scene information (entities, relationships). This study's visual information extraction module combines object detection and pose

estimation, providing a more effective representation of scene information than methods relying on either approach alone. Additionally, a scene graph is a computable structure that intuitively represents entities and their relationships in a rich and organized way. Xiong et al. [5] developed a scene perception framework for multi-class PPE detection. They used OpenPose to estimate worker poses, integrated the detected 2D key points as spatial anchors to identify part attention regions, and employed a CNN-based classifier to determine whether PPE in those regions was correctly used. This study leverages the inherent relationships between PPE and local body parts (such as head-helmet and upper body-vest) to determine whether PPE is being worn correctly, without requiring a detailed analysis of the spatial relationships between the objects. In 2022, Kim et al. [17] proposed a semantic graph neural network approach that represented construction resources and their interactions as graphs. They used a neural message-passing mechanism to simulate propagation effects within the graph. In this study, visual relationship detection focuses on three key features: spatial, visual, and operational context. The construction scene graph generation model proposed by Zhang et al. [7] uses a transformer-based network to capture entities and relationships and format them into a construction scene graph. In 2024, Kim et al. [63] detects four types of relationships between workers and heavy construction machinery (guiding, adjusting, filling, and not working with) by employing a DNN architecture, Pixel2Graph. Xie et al. [64] explores 3D human-object interaction (HOI) detection on construction sites through a novel context-guided graph neural network, integrating transferable contextual knowledge and optimizing interaction detection via bipartite matching loss. The



Fig. 8. Summary of research on relationship-based perception.

summary of research on relationship-based perception is shown in Fig. 8.

3.1.3. Event-based perception

3.1.3.1. Single event. An event is a perceptible, dynamic occurrence that exists independently and is inherently non-static. Typically, an event involves a sequence of actions or a series of state changes, representing a specific fact that evolves over time. In cognitive science, Zacks [65] of Stanford University proposed that events are behaviors observed by bystanders in the real world and can be understood through spatio-temporal structures. In construction, researchers primarily focus on events that could lead to safety accidents, such as collisions, slips, and falls from heights, as shown in Fig. 9.

In 2016, Wang et al. [66] introduced two innovative four-dimensional models, the time-sphere model and the time-cuboid model, to effectively prevent collisions between workers and equipment or between pieces of equipment during construction. The four-dimensional model analyzes time-series sensor data to capture the state of entities, including 3D position, direction, and speed. It uses a sphere or cuboid to represent the hazardous area around equipment or workers. The warning distance is adjusted and updated according to the collected entity state information, and collision accidents are identified and predicted based on information such as relative position, movement direction, and speed. In 2017, to address collision accidents in road construction, Park et al. [67] developed and tested a BLE-based short-range sensing and alarm system to overcome inconsistencies and delays in proximity warning systems. In 2019, Jo et al. [68] used ultra-wideband (UWB) sensing technologies to identify construction resources, perceive collision accidents, and provide real-time warnings. In 2021, Golovina et al. [69] studied data-driven approaches for managing close-call incidents on construction sites and enhancing event response. Shakerian et al. [70] proposed a method based on wearable biosensors and machine learning to perceive the heat strain risk of workers on construction sites. In 2022, Fang et al. [71] developed a content-based image retrieval approach (CBIR) that fused four features (deep learning, color, texture, and shape-based features) with certain weights to retrieve images containing workers' unsafe behaviors. In 2024, Mei et al. [72] combined an environment-based detection module using Mask-R-CNN with a posture-based detection module employing skeleton detection. They also introduced a data fusion module to automate the detection of guardrail climbing behavior on construction sites.

3.1.3.2. Event classification. Previous research focused on individual hazardous events rather than a variety of potential risks. For example, studies have examined issues such as workers not wearing PPE, scattered screws on construction sites, or undocumented workers performing specific tasks. However, the construction scene is a complex collection. Focusing on a single event is insufficient. To achieve comprehensive scene understanding, it is essential to perceive multiple events simultaneously. The summary of research on event classification is shown in Fig. 10.

In 2018, Antwi-Afari et al. [73,74] developed a supervised machine learning method to detect and classify normal walking and four loss of balance events (slip, trip, unexpected step-down, twisted ankle) using plantar pressure distribution data from wearable insole pressure sensors. In 2019, Cai et al. [8] used a multi-classification model based on LSTM to identify specific activities of a workgroup on construction sites, including road paving, spotting, and others. Zhang et al. [75] used accelerometers and gyroscopes embedded in smartphones to collect three-axis acceleration and angle. Then, they applied artificial neural networks to analyze these data and distinguish normal walking from three near-miss falls: (1) slip and trip, (2) sudden body sway, arm swing, and unsteady footsteps, and (3) entering dangerous zones. In 2020, Li et al. [76,77] used a wearable insole pressure system to measure subjects' gait patterns. They then calculated five gait variability parameters to distinguish between normal gait, slip hazard, trip hazard, and unexpected stepdown hazard. Chen et al. [9] proposed a semantic region detection method with transfer learning and saliency calculation. This method detects semantic regions of different levels, classifying them into primary activity-active region, secondary activity-active region, materials and tools region, and main component region. Semantic regions highlight the areas that contain the most relevant information, revealing key objects and events. This new method can be used to represent event regions (such as construction activity), crop image size, reduce storage space, and add semantic annotations for improving image retrieval. Lee et al. [78] developed an audio-based event detection system that used K-Nearest Neighbor (KNN) to classify sounds in construction. The system identified ten construction activities: excavating, concrete mixing, compacting, bulldozing, hammering, piling, concrete pumping, drilling, and operating a crane and a chainsaw. In 2021, Jeelani et al. [79] utilized visual SLAM and semantic segmentation with Mask R-CNN to detect worker proximity to six pre-defined static hazards and six dynamic hazards. In 2022, Lee et al. [80] used a waist-worn IMU sensor to record waist movement and applied the Mahalanobis distance (MD) metric to detect slip, trip, and fall (STF) events, including risky

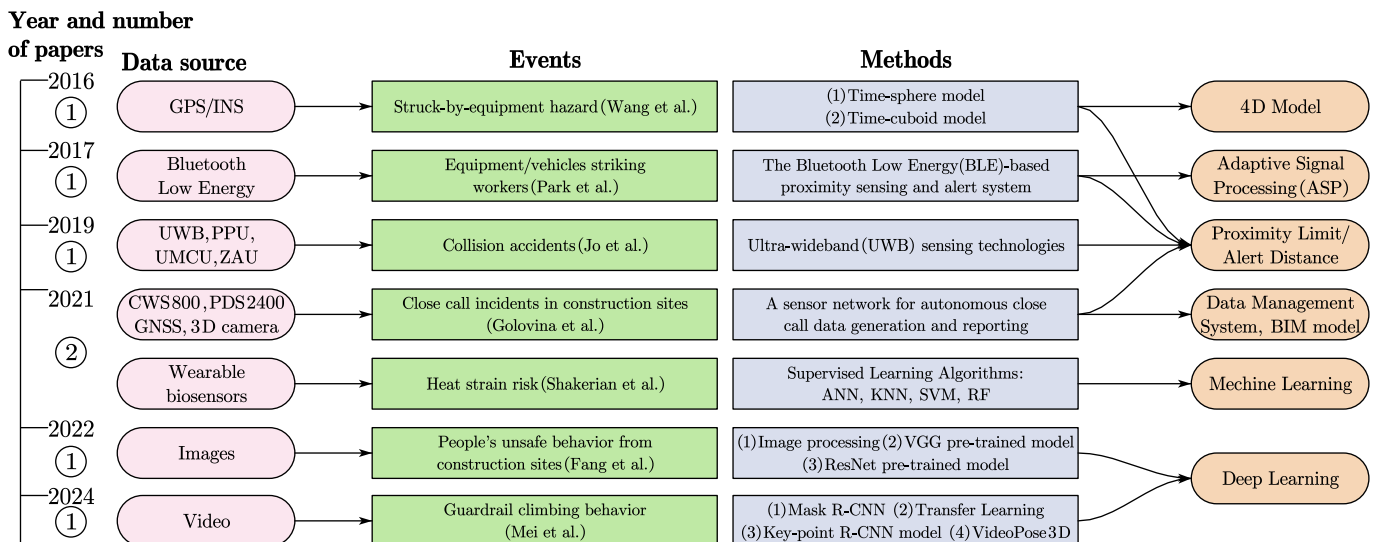


Fig. 9. Summary of research on single-event perception.

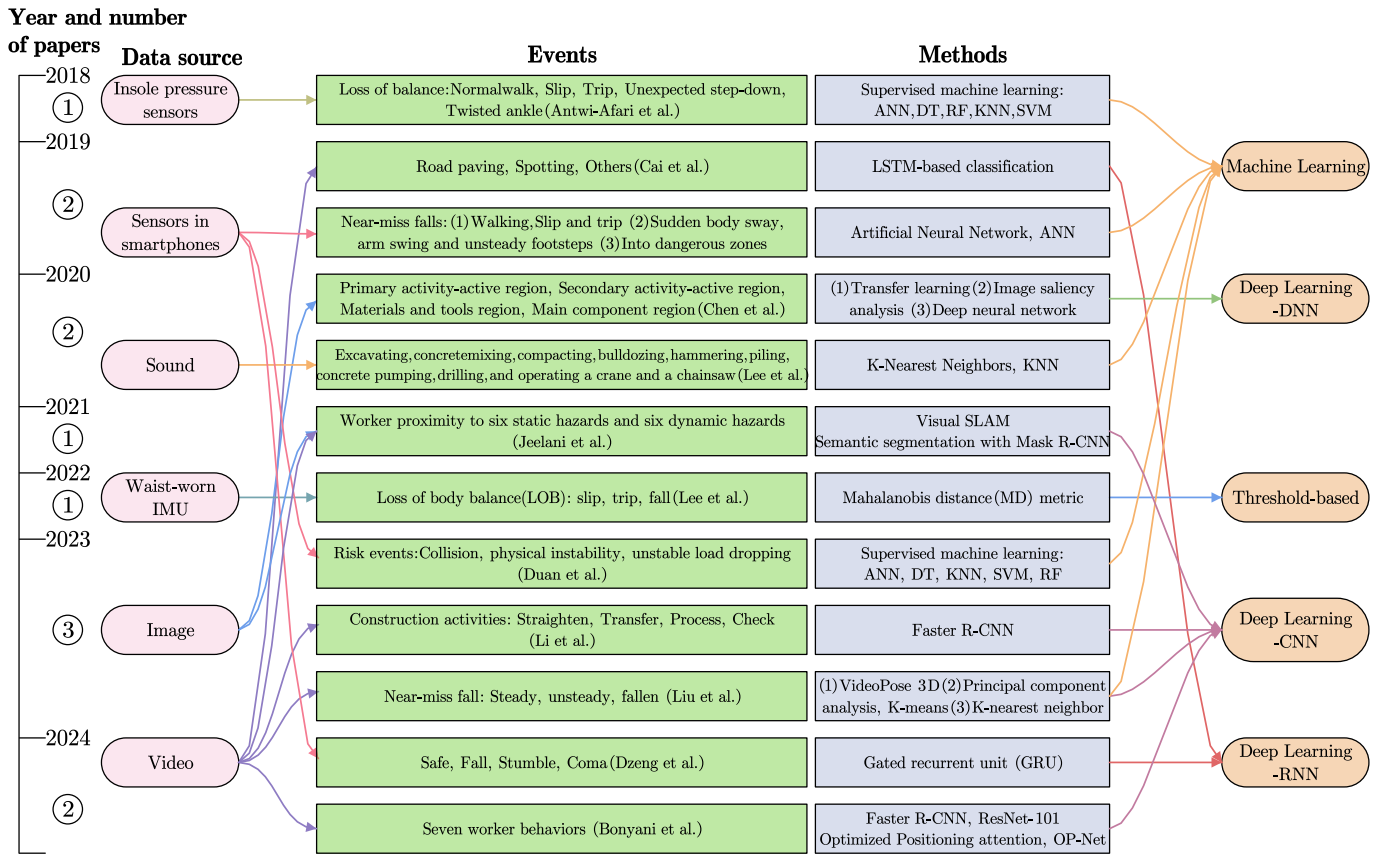


Fig. 10. Summary of research on event classification.

movement, inattentive walking, and risky environment. In 2023, Duan et al. [81] proposed a method for two material handling tasks, which used built-in sensors in smartphones to collect acceleration and angular velocity. They then used supervised machine learning to identify and classify four types of risk events, including collision, physical instability, unstable load, and dropping. Li et al. [12] used on-site video and Faster R-CNN to identify four construction activities (straighten, transfer, process, and check) in the steel processing scene. Liu et al. [82] proposed a method for detecting near-miss falls using computer vision and machine learning. First, VideoPose3D was used to extract the 3D coordinate matrix of workers' joints from 2D RGB images, and then Principal Component Analysis (PCA) was used to downscale the high-dimensional matrix. Finally, KNN was used to divide the worker's pose state into three stages: steady, unsteady, and falling. In 2024, Dzeng et al. [83] developed a new algorithm based on recurrent neural networks, which outperformed the existing hierarchical threshold model in detecting stumble and fall events. The new algorithm demonstrated higher sensitivity, lower false alarm rates, and a significant reduction in false alarms for coma events. Bonyani et al. [84] introduced the Optimized Positioning (OP-Net) architectures and an attention-based spatiotemporal sampling approach to detect seven worker behaviors in the CMA dataset.

3.1.4. Image captioning

Before 2020, research primarily focused on identifying individual elements (such as objects, attributes, and relationships) from visual signals or detecting single events (such as falls, and collisions) from sensor signals. These methods can automatically identify objects, relationships, and events, but they fail to extract deeper information about construction scenes, such as quantities, environmental factors, and other relevant details. Only a few studies consider the construction scene as a whole, cohesively incorporating deep semantic information (such as

multiple objects, relationships, attributes, and environmental factors). Image captioning is a multimodal task that combines computer vision and natural language processing, allowing computers to "speak for themselves by looking at pictures". For image captioning, the computer must not only identify elements in images but also describe them in natural language. By fusing text and visual features, the computer can extract rich contextual information from the visual background.

In 2020, Kim et al. [16] proposed a method for generating spatiotemporal and visual contextual information from image data collected by drones. This method combines Faster R-CNN, soft attention mechanism, and long short-term memory (LSTM) to generate textual information about location, status, motion, color, and quantity of construction resources. Liu et al. [85] proposed an image captioning method, integrating a CNN encoder (VGG-16 or ResNet-50) for image representation and an RNN decoder (LSTM) for sentence generation. The model was established to describe five construction scenes: cart transportation, masonry work, rebar work, plastering, and tiling. In 2022, Liu et al. [86] developed a visual grounding approach to detect and locate six unsafe behaviors. This study uses pre-trained YOLOv3 as image encoder to extract visual features and BERT as a text encoder to extract text features. It also employs recursive sub-query to extract contextual information related to unsafe behavior. Wang et al. [87] proposed an integrated information extraction method to extract semantic information such as objects, activities, and interactions in construction images. They used Mask R-CNN as encoder and two-layer LSTM with attention units as decoder, namely Attention LSTM and Language LSTM. Xiao et al. [88] explored the feasibility of image captioning in construction, selected six deep learning algorithms, and tested them on a construction captioning dataset. In 2023, Zhai et al. [2] developed an attention-based image captioning model to extract worker-centric semantic information in construction scenes. Zhong et al. [89] proposed an image captioning model for identifying semantic

information such as multiple objects, relationships, and attributes from construction videos. They integrated the ResNet101 encoder with the LSTM-Attention decoder and tested the model in five construction scenes: formwork, masonry, rebar work, cart transport, and plastering. In 2024, Chen et al. [14] further developed a Visual Construction Safety Query (VCSQ) system based on image captioning for construction scenes. Song et al. [90] proposed a style-enhanced Transformer model for image captioning based on the characteristics of construction scenes. This study not only used a semantic encoder to extract image semantic information but also used a style encoder to extract style information. Additionally, a sentence-style loss was incorporated into the total loss function. Gil et al. [91] utilized the ClipCap model to generate image captions for cropped body images of construction workers. The summary of research on image captioning is shown in Fig. 11.

Visual Language models combine visual and textual information to process and interpret multimodal data. Therefore, they can effectively analyze images and generate natural language descriptions, providing a more accurate representation of complex relationships between dynamic objects in the construction environment.

3.2. Reasoning ability

The research mentioned above focuses solely on perception ability, specifically the identification of semantic information such as objects, attributes, relationships, and events. However, the ultimate goal of scene understanding is to make inferences and decisions based on this perceived information, which requires integrating domain-specific rules and knowledge from the construction field. With the development of knowledge engineering, rule reasoning in construction has also experienced the development process: simple rule-based reasoning, ontology-based reasoning, scene graph-based reasoning, and NLP-based reasoning.

3.2.1. Simple rule-based reasoning

Simple rules are mainly defined by humans. Inference rules based on expert knowledge are then used to evaluate key management indicators

in construction, thus improving management of safety, productivity, and progress. There are various technical means, such as expert systems, computer programs, and 3D or 4D building information systems (BIM).

In 2020, Zhang et al. [61] used fuzzy inference to conduct worker safety assessments on construction sites. The input of the fuzzy inference system is the proximity and crowding of workers, and the output is the safety level of workers. In 2021, Li et al. [10] determined the number of workers involved in a task by analyzing the positional relationship between detected objects and workers. They then calculated the production efficiency of reinforcement binding based on the number of workers and the duration of the task. In 2022, Xue et al. [13] registered the detected building components into BIM, and inferred construction progress by calculating the number of built components. In 2023, Chen et al. [25] proposed a geometric relationship analysis algorithm for analyzing stairway safety on construction sites. Based on stairway safety regulations, which state “always use handrails when going up and down stairs,” the study defines a worker as safe if their wrist keypoints are within the handrail’s detection box. If not, the situation is flagged as dangerous. Using YOLOv4 to detect entities like workers, materials, and equipment, Lee et al. [40] set four rules and developed a fixed computer program to assess whether a dangerous situation exists. Li et al. [12] calculated the overlap rate between worker detection boxes and activity detection boxes to identify each worker’s construction activities, thus monitoring their work status. Based on detected entities such as workers, PPE, and portable ladders, Park et al. [41] set corresponding safety rules according to different working heights. Then, they developed corresponding hazard detection algorithms based on safety rules to monitor workers’ safety status when using portable ladders. Seong et al. [42] defined “danger areas” and proposed a method to assess collision risk based on the position and speed of mobile equipment. Tran et al. [43] defined danger zones in 4D BIM based on predefined safety rules and transferred these danger zones to the first frame image captured by the camera. Then, the YOLOv8-based model reported an unsafe event whenever it detected a worker entering the danger zone. Wójcik et al. [47] developed an algorithm to detect helmet compliance, which analyzes the spatial relationship between the helmet and the head’s center

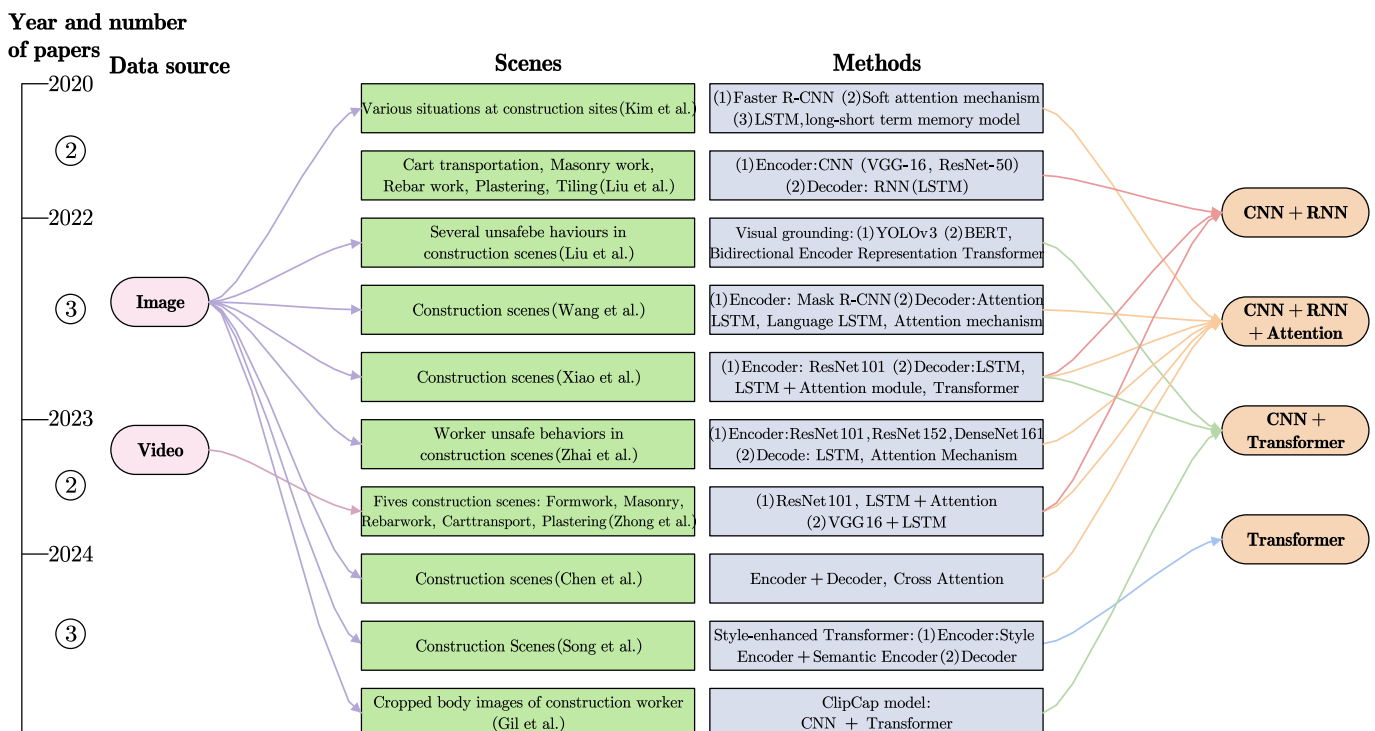


Fig. 11. Summary of research on image captioning.

to determine if the helmet is worn correctly. This study uses a simple, interpretable rule-based algorithm, avoiding the intraclass similarity issue on wearer/non-wearer detection. Zhao et al. [46] categorized hazardous areas into three types: civil construction hazards, machinery movement hazards, and power leakage hazards. They also defined the attribute information of workers. If the CNN-based object detection model detected a worker without required attributes entering a hazardous area, it was flagged as unsafe. In 2024, Yang et al. [92] assessed the proper use of PPE by analyzing the positions of PPE on workers and the corresponding coordinates of workers' keypoints. The summary of research on simple rule-based reasoning is shown in Fig. 12.

It can be found that simple rule-based reasoning is only applicable to a specific scenario. It has advantages such as simplicity and strong interpretability, as well as defects such as narrow application range and poor robustness. It struggles to handle the complexity, dynamics, and variability of construction scenes effectively.

3.2.2. Ontology-based reasoning

Previous studies have significantly improved the accuracy and efficiency of object detection and relationship recognition. However, these tasks remain focused on low-level information extraction, making it challenging to automatically detect and identify complex events solely through feature extraction and image processing. Construction hazardous events involve high-level semantic information that deep learning algorithms alone cannot extract from images. Therefore, a method capable of delivering robust semantic understanding is essential. Ontology is a knowledge representation structure that offers unambiguous, standardized terminology, enabling effective communication among all participants in the reasoning process [93]. Ontology can formalize concepts and their relationships in a specific domain [94]. Using ontology as a tool for event description helps bridge the gap between low-level and high-level semantics [95,96]. For complex and

diverse construction scenes, combining ontology with deep learning is particularly valuable for enhancing scene understanding.

In 2019, Xiong et al. [24] developed an ontology model for construction safety. This model was used to convert visual triples extracted from images and safety rules from documents into logical clauses. They then built an inference engine using the Prolog programming language to automatically identify potential risks in construction scenes. In 2020, Zhang et al. [62] developed an ontology semantic network for foundation pit excavation scenes. They used the ontology's logical language to translate low-level image semantics into high-level event descriptions and converted construction risk rules into ontology rules. Finally, they employed the Pellet inference engine to detect potential high-risk situations on construction sites. In 2022, Shen et al. [97] proposed an automatic safety risk identification and prevention mechanism for the construction process, integrating a safety rule library based on ontology and natural language processing. In 2024, Johansen et al. [98] used an ontology to formalize the components and attributes of guardrail systems as logical facts and established rules to evaluate their compliance. The summary of research on ontology-based reasoning is shown in Fig. 13.

Manually extracting regulatory documents and encoding them into a computer-readable format is time-consuming, costly, and prone to errors. Additionally, for more complex and diverse construction scenes, establishing a more detailed ontology semantic network and expanding the corresponding rule base is essential. However, this lack of scalability significantly limits the method's practicality in engineering.

3.2.3. Scene graph-based reasoning

The scene graph, defined by Johnson et al. [99], is a computable graph structure that intuitively represents numerous entities and their relationships. In this graph, nodes represent entities, while edges denote the pairwise relationship. Computer vision is used to automatically

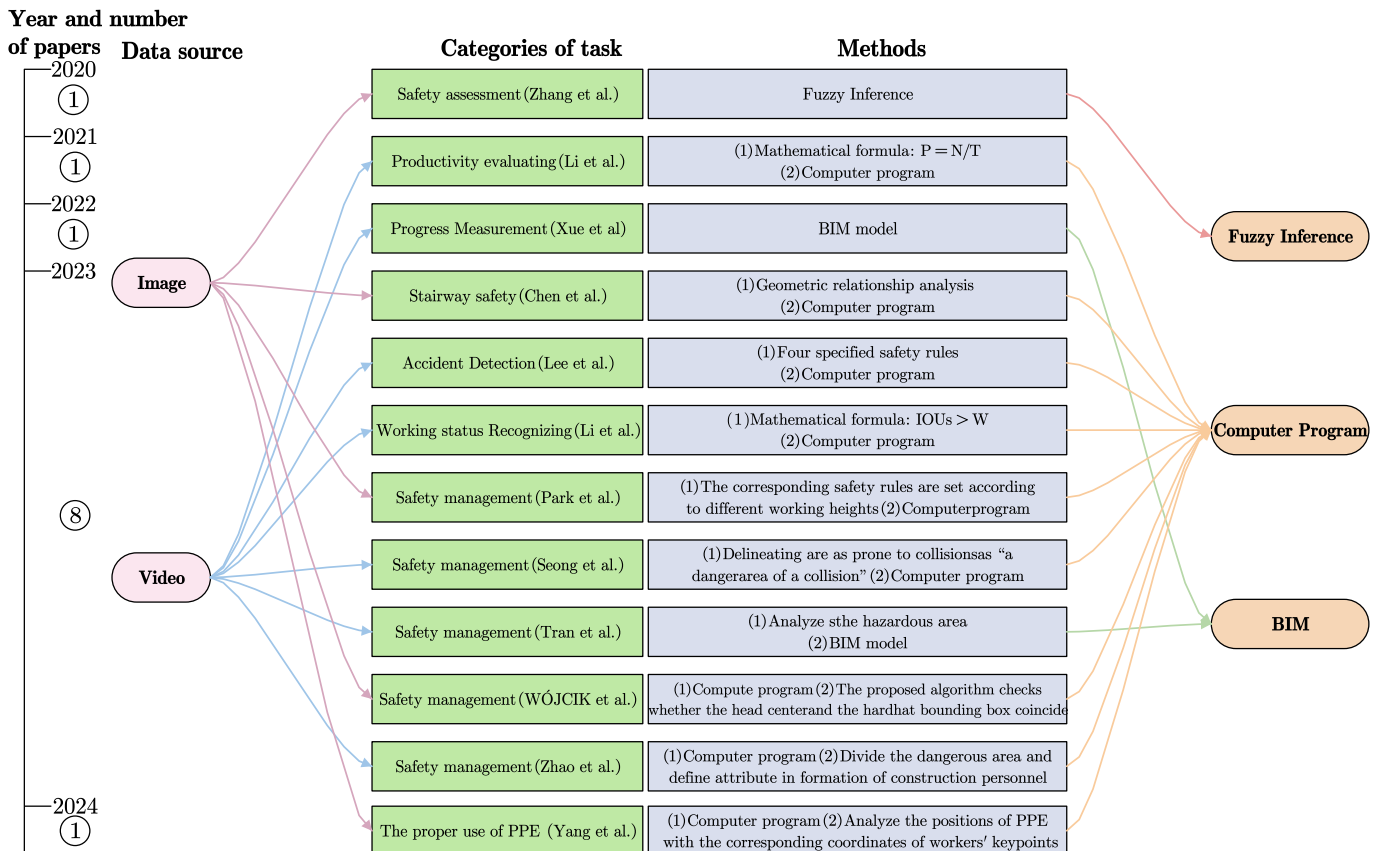


Fig. 12. Summary of research on simple rule-based reasoning.

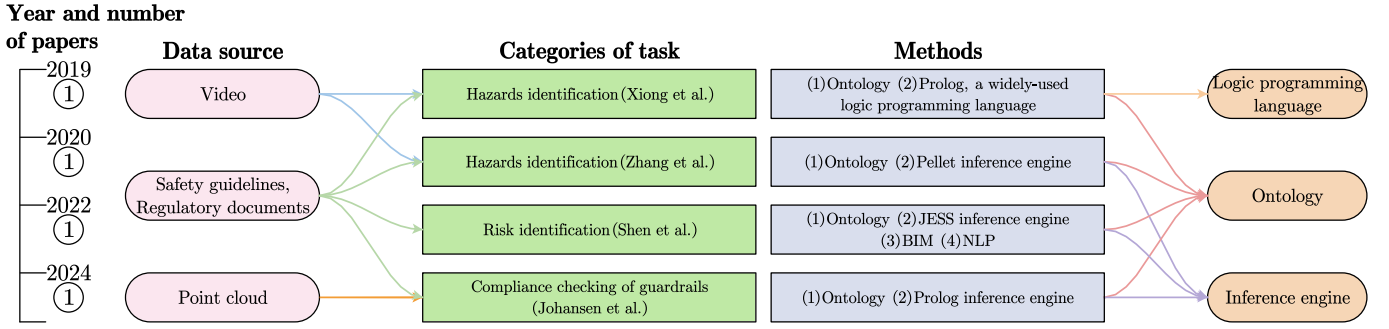


Fig. 13. Summary of research on ontology-based reasoning.

extract entities from images and analyze their relationships, transforming objective visual information into graph structures for a more effective representation of scene details. Additionally, by generating syntactic features and performing semantic analysis, construction rule texts can be automatically processed, transforming natural language sentences into computable graph structures. Furthermore, reasoning analysis can be conducted by comparing the scene graph of on-site images with the scene graph of relevant regulatory rules to perform compliance checks.

In 2021, to address the limitation of traditional scene graphs in representing conditional scenarios (such as the use of safety belts when working at heights), Chen et al. [3] proposed a hierarchical scene graph structure for automatic reasoning about improper PPE usage. In the visual information extraction module, entity detection and relationship analysis are conducted on on-site images, with a focus on the worker. The resulting visual semantic triples are then converted into computable scene graphs. In the linguistic information extraction module, a method is proposed to represent and convert text information by encoding regulatory rules into a hierarchical scene graph. Safety rules are decomposed into semantic phrases and represented as triples (object 1, relationship, object 2). To capture the conditional relationships between the semantic phrases, the triples are classified into conditional triplets and instructed triplets. A hierarchical scene graph of regulatory rules is then generated based on these semantic phrase triples. In the automatic reasoning module, the scene graphs generated by the linguistic and visual information extraction modules are used as input to analyze and reason about dangerous situations through graph structure analysis. However, due to the vast number of regulatory documents, inconsistent formats, and the complexity of semantic information, manually processing these rules is time-consuming, costly, and prone to errors. In 2022, based on the previous research [3], Chen et al. [4] improved the linguistic information extraction module and proposed a semantic analysis method based on natural language processing (NLP) and ontology. Through syntactic feature generation and semantic analysis, linguistic information is automatically extracted from regulatory rules, converting natural language sentences into computable graph structures. The summary of research on scene graph-based reasoning is shown in Fig. 14.

Ontologies and scene graphs share conceptual similarities, yet their core emphases differ. From the perspective of definition and purpose,

ontologies are formal specifications of a conceptual domain, structured to systematically represent shared knowledge in a reusable manner [100]. They consist of concepts, relationships, and axioms, enabling reasoning and interoperability across applications. For instance, ontologies have been utilized in construction safety planning to integrate safety knowledge with Building Information Models (BIM) [93] and in knowledge retrieval systems to enhance information discovery [94]. In contrast, scene graphs are graph-based representations that describe the spatial and semantic structure of a scene in terms of objects, attributes, and relationships, primarily supporting tasks like image retrieval and scene understanding [99].

From the perspective of structure and scope, ontologies emphasize hierarchical structures and logical constraints, enabling domain-independent reasoning and rule-checking. For example, Zhang et al. [93] demonstrated the use of ontologies to automate hazard identification by integrating safety information within BIM. Scene graphs, however, are inherently visual and domain-specific, focusing on object-centric details to provide granular descriptions of visual data [99].

Despite their differing origins, ontologies and scene graphs overlap in their graph-like structures and ability to represent relationships. Ontologies can enhance scene graphs by formalizing domain knowledge, thereby enriching their semantic and inferential capabilities. For instance, embedding an ontology within a scene graph framework can standardize object attributes and contextual relationships, as seen in ontology developed by Zhang et al. [93] for safety planning, which visualizes safety information through BIM. By leveraging their complementary strengths, integrating ontologies with scene graphs offers significant potential for improving reasoning and retrieval in both visual and non-visual domains.

3.2.4. NLP-based reasoning

The goal of natural language processing (NLP) is to achieve effective communication between humans and computers, allowing computers to understand natural language and express intentions in natural language. Generally, NLP involves two main aspects: natural language understanding and natural language generation. Its research spans multiple levels of granularity, including characters, words, phrases, sentences, paragraphs, and chapters. With the advancement of natural language processing and large-scale pre-trained corpus models, NLP-based methods have attracted significant attention from researchers in

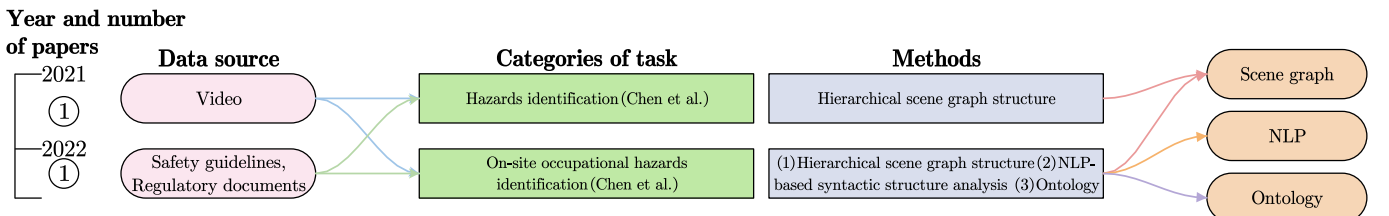


Fig. 14. Summary of research on scene graph-based reasoning.

construction and hold great application potential [101]. However, research on using NLP to understand domain knowledge to enhance construction scene understanding is still limited. In 2022, Zhang et al. [7] developed a hazard inference module based on the pre-trained language model C-BERT. Visual facts from the construction scene graph model, along with external domain knowledge such as construction regulations, are combined into a unified vector representation and encoded by C-BERT for hazard inference in construction. Specially, Liu et al. [86] combined computer vision and natural language processing to develop an end-to-end model for detecting and locating unsafe behaviors on construction sites. This model integrates the perception ability and reasoning ability of scene understanding, which is more conducive to global optimization. In 2024, Chen et al. [14] developed a visual construction safety query (VCSQ) system by integrating image captioning (IC), safety-centric visual question answering (VQA), and keyword-based image-text retrieval (ITR). The system utilizes pre-trained visual object detectors and large language models (LLMs) to explore new applications in construction safety, serving as a valuable tool for enhancing on-site situational awareness. Gil et al. [91] utilizes cosine similarity to compare text embeddings of automatically generated image captions with predefined prompts based on OSHA PPE regulations, evaluating workers' compliance with PPE. Wen et al. [102] employed a visual question answering (VQA) model to verify PPE compliance and generate safety inspection reports. The summary of research on NLP-based reasoning is shown in Fig. 15.

Natural language processing and large language models have made significant progress over time. In 2020, OpenAI released GPT-3, a generative pre-trained language model with 175 billion parameters. Subsequently, Google developed the Switch Transformer, the first model to exceed one trillion parameters. These large language models, often called "large models," have demonstrated exceptional performance in text generation, few-shot learning, zero-shot learning, and inference tasks. The release of ChatGPT and GPT-4 has propelled large language model (LLM) research into large-scale applications, sparking a new wave of advancements in natural language processing. For construction scene understanding, research leveraging NLP focuses on converting rules and expert knowledge stored in natural language into computer-readable information. This transformation enhances the perception and reasoning abilities of construction scene understanding. Recently, the rise of pre-trained LLM has opened new avenues for advancing construction scene understanding.

4. Analysis and research trends

4.1. From application-driven neural networks to general artificial intelligence in construction

Currently, artificial intelligence algorithms in construction are designed for specific tasks and application scenarios. However, due to the data dependency of deep learning, creating training datasets and fine-tuning model parameters for specific scenarios and applications

incur high development costs. Moreover, both visual perception models and language reasoning models have limited generalization and transfer capabilities, making it challenging to quickly adapt them to new application scenarios.

According to Microsoft Research, GPT-4 excels in answering questions across diverse fields, including mathematics, programming, vision, medicine, law, and psychology, without requiring specific design instructions. Its performance significantly surpasses ChatGPT and approaches human-level capabilities, making it a reasonable candidate for an early, albeit incomplete, general artificial intelligence system [103].

The emergence of ChatGPT demonstrates that a general generative architecture can handle a wide range of natural language processing tasks. Compared to task-specific architectures, general architectures leverage task diversity, enable efficient knowledge sharing across tasks, and utilize training data more effectively. This shift significantly impacts the traditional paradigm of task-specific architectures. Based on the above research, developing general artificial intelligence systems tailored to the construction field is expected to become a future research trend, as illustrated in Fig. 16.

4.2. Perception from single object and relationship to comprehensive objects and relationships

Research on construction scene understanding has evolved from perceiving entities to perceiving relationships, from single objects to complex ones, and from single relationships to complex ones, as illustrated in Fig. 17.

For example, in 2015, Park et al. [58] determined whether a worker was wearing a helmet correctly by analyzing the geometric and spatial relationship between the worker and the helmet. As an early exploration of computer vision in construction, this study has some limitations. First, it used a template to detect workers, which only allowed detection of standing workers and failed to identify workers in other postures, such as squatting, bending over, or sitting. Second, this method relies on the spatial and geometric relationship between the worker and the helmet to match them, using specific parameters that only work for standing workers. If the worker adopts a different posture, these parameters need to be adjusted accordingly. Early studies used template-based approaches to detect entities and their relationships in construction scenes. However, these methods are sensitive to scale and rotation, requiring a redesign of the template or matching rules each time a type of entity or relationship is identified. This limitation makes it difficult to detect complex entities and relationships.

With the rise of deep learning, research on construction scene understanding has progressed significantly. In 2022, Kim et al. [17] introduced a semantic graph neural network (semantic GNN) to classify visual relationships between multiple resources, covering thirteen relationship types. Their research discovers underlying domain-specific natures of visual relationships (semantic features and propagation effects) and integrates various semantic features, including spatial, visual, and operational contexts. Unlike methods relying on if-then rules, this

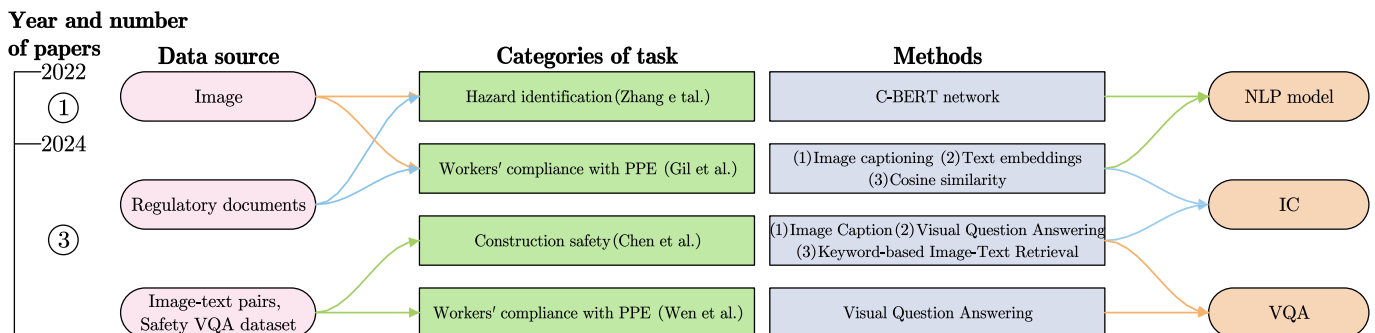


Fig. 15. Summary of research on NLP-based reasoning.

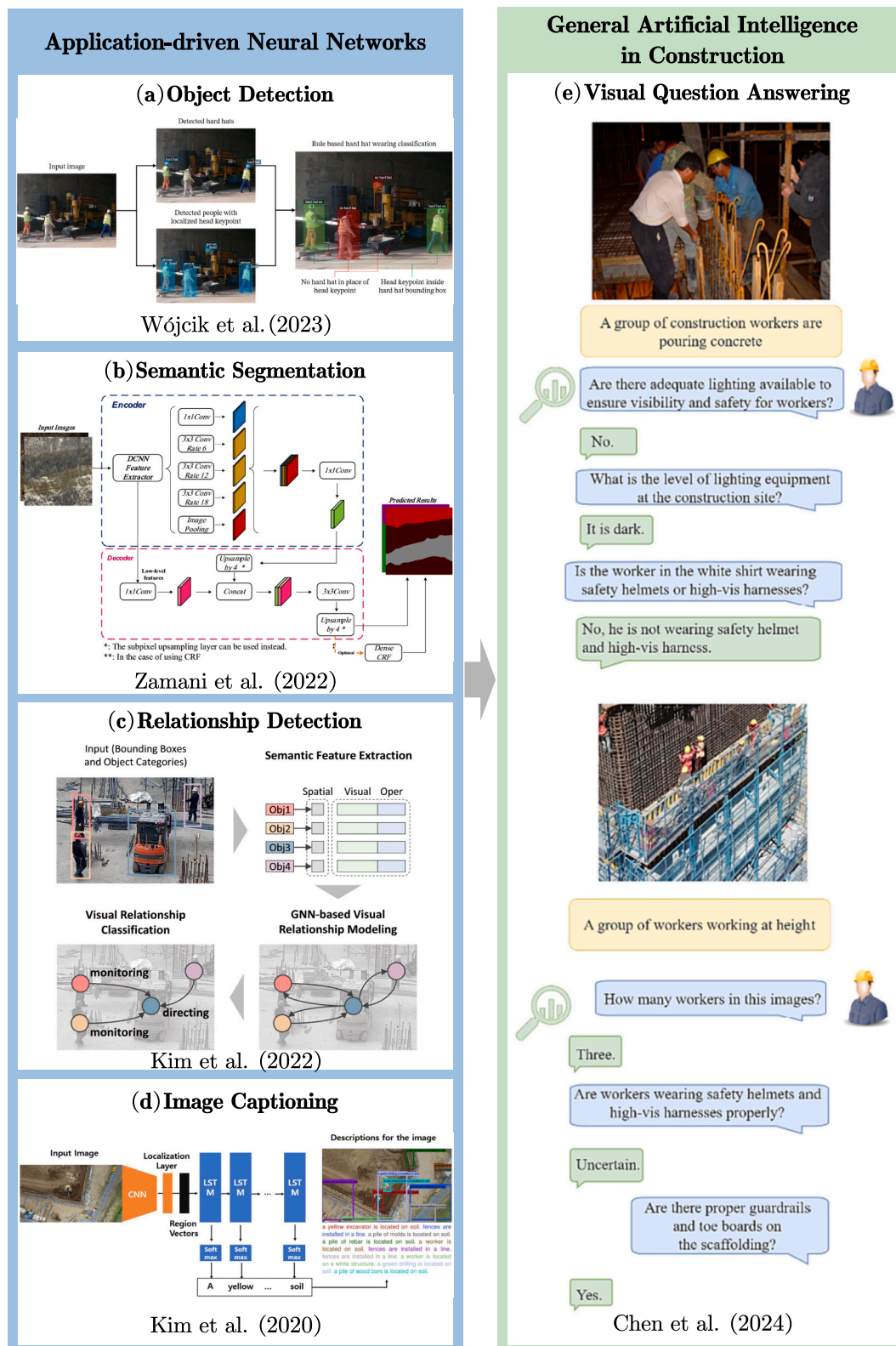


Fig. 16. Evolution from application-driven neural networks to general artificial intelligence in construction ((a)Object Detection [47]; (b)Semantic Segmentation [18]; (c)Relationship Detection [17]; (d)Image Captioning [16]; (e) Visual Question Answering [14]).

approach offers greater scalability and applicability. Given the appropriate training data, it can be easily extended to classify various types of visual relationships.

Construction scenes contain numerous objects that interact in complex ways. To fully understand these scenes, it is essential to recognize

both the construction resources and their spatial and semantic relationships. Describing the scene using various elements such as objects, relationships, and attributes is key to capturing the complexity of construction environments.



Fig. 17. Evolution of perception: from single object and relationship to comprehensive objects and relationships ((a)Worker [46]; (b)Excavator [104]; (c)Helmet [105]; (d)Vehicle [104]; (e)Welding [7]; (f)Scaffolding [7]; (g)Excavating [7]; (h)Bricking [7]; (i)Between workers and vehicles [6]; (j)Between workers and handrails [25]; (k)Between workers and structural supports [59]; (l)One-to-one interactions [17]; (m)One-to-many interactions [17]; (n)Many-to-many interactions [17]).

4.3. From rule-based learning to end-to-end learning

For the perception ability of scene understanding, early studies relied on manual feature extraction methods, which often overlooked the rich

semantic relationships between multiple objects. This limitation affected the accuracy, robustness, and scalability of scene understanding. Since these models were designed based on specific rules, the algorithms performed well in scenes that followed those rules. However,

the model's effectiveness was limited in more diverse and complex scenarios, especially when it came to transferring to other construction scenes. Additionally, the manual feature extraction process is time-consuming, costly, and prone to errors. For the reasoning ability of scene understanding, the reasoning module often relies on third-party software (such as fuzzy inference [61], ontology reasoning engine [62]) or adopts additional natural language processing models (such as C-BERT [7]) to extract rules. These factors affect the practicality and scalability of these methods.

End-to-end frameworks have the potential to greatly improve the practicality of scene understanding. While these approaches are challenging, they offer several key advantages. First, they can fully leverage the data, eliminating the need for expert knowledge, as long as enough data is available. Second, they require fewer manually designed components, simplifying the design workflow. Researchers can focus less on manually designing functions or intermediate representations. Third, end-to-end neural network training reduces error propagation, which can enhance the performance of downstream tasks. Fig. 19 illustrates the learning paradigms from rule-based learning to end-to-end learning.

For the perception ability of scene understanding, related research has evolved from rule-based methods [22,34,58] to end-to-end learning [18,35], as illustrated in Fig. 18. For instance, in 2016, Kim et al. [22] applied Nearest Neighbors and Scale Invariant Feature Transform Flow Matching for segmentation. The method retrieved relevant images and passed the label to the query image to segment various objects on construction sites. However, it relies on having similar images in the database. If the database lacks construction images similar to the query image, the accuracy of segmentation will significantly decrease. This is particularly challenging for construction scenes, which are diverse, complex, and constantly changing, with a wide range of workers, materials, and machinery. Therefore, even within the same type of construction project, it is challenging to find similar scenes. This characteristic limits the effectiveness of segmentation using this method. In 2020, Atkinson et al. [35] used Mask-RCNN to segment enclosed space scenes under the floor. In 2022, Zamani et al. [18] employed Deeplabv3+ to segment various soils in the scene. This end-to-end segmentation approach prioritizes robustness, even at the cost of

reducing the interpretability of the framework. Previously, researchers spent significant time on feature extraction and feature engineering for specific machine learning tasks, but deep neural networks, supported by large labeled datasets, have introduced a new paradigm for these tasks. For natural language processing tasks, in 2021, Chen et al. [3] used a manual method to extract information from regulatory documents and generate a hierarchical scene graph. However, this process is time-consuming and prone to errors, especially with large volumes of regulatory documents. In 2022, Chen et al. [4] improved the process by using an NLP and ontology-based approach, which automatically processes input regulatory rules and converts natural language sentences into computable graph structure representations.

For the reasoning ability of scene understanding, most existing studies still require a separate reasoning module based on perception to achieve reasoning ability [3,4,7,24,61,62]. However, as an exception, Liu et al. [86] took a different approach by unifying perception and reasoning abilities. They used an end-to-end visual grounding method to identify unsafe behaviors on construction sites. For example, in 2020, Cao et al. [61] used fuzzy inference to assess the safety level of construction scenes by analyzing proximity and crowdedness within the scene. Zhu et al. [62] developed a semantic web rule language (SWRL) rule base and utilized the Pellet inference engine to assess risks in construction scenes. These studies aim to combine ontologies to create semantic models and leverage rules from text to enhance the understanding of unsafe behaviors [24,62]. While unsafe behaviors can be identified from images, they cannot be intuitively located in images. In contrast, in 2022, Liu et al. [86] introduced an end-to-end, single-stage visual grounding method to identify and locate multiple unsafe behaviors in images based on query text related to unsafe behaviors. This end-to-end model reduces error propagation and enhances the overall performance of the task.

4.4. From data dependency to transfer learning and self-supervised learning

Deep learning has significantly advanced construction scene understanding, but gathering training data for diverse scenes and tasks

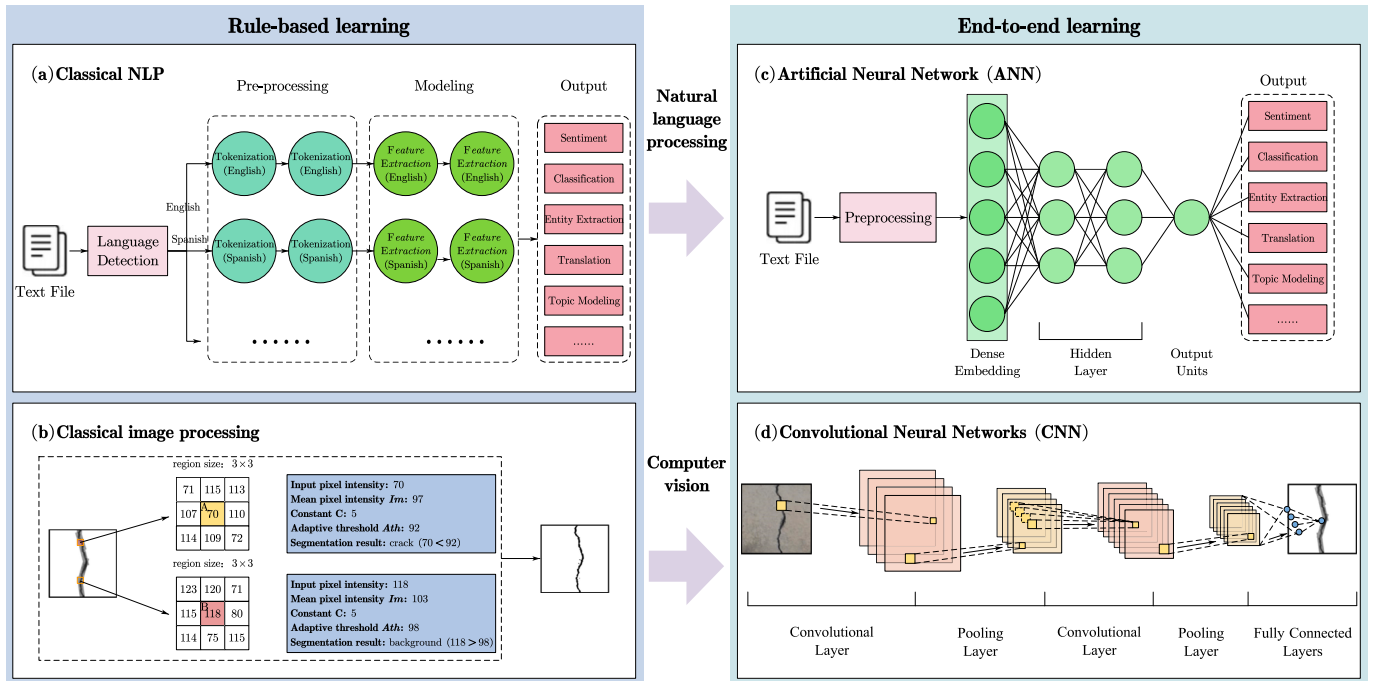


Fig. 18. Evolution from rule-based learning to end-to-end learning ((a)Classical NLP; (b)Classical image processing [106]; (c)Artificial Neural Network (ANN); (d) Convolutional Neural Networks (CNN)).

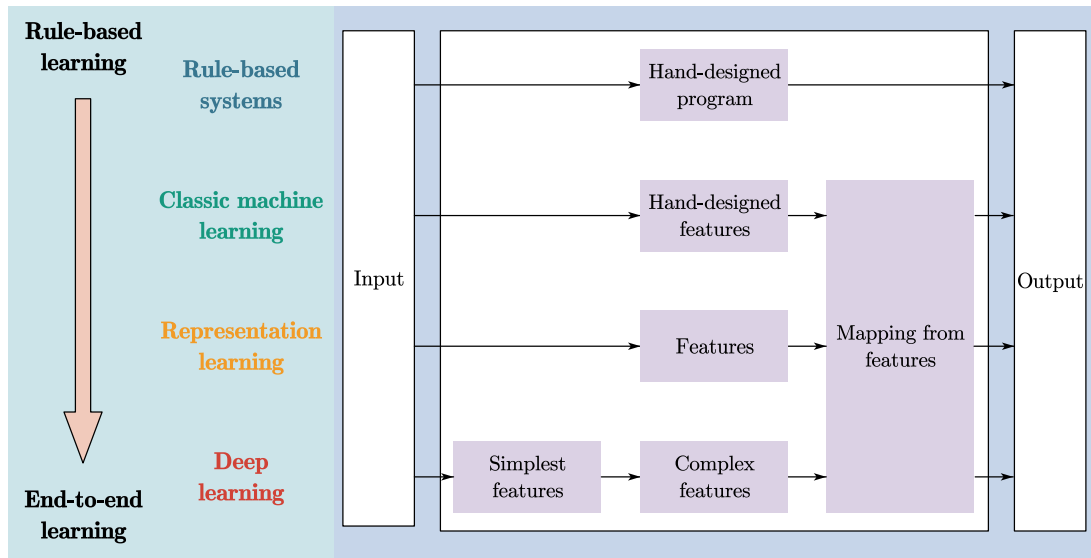


Fig. 19. Learning paradigms: from rule-based learning to end-to-end learning.

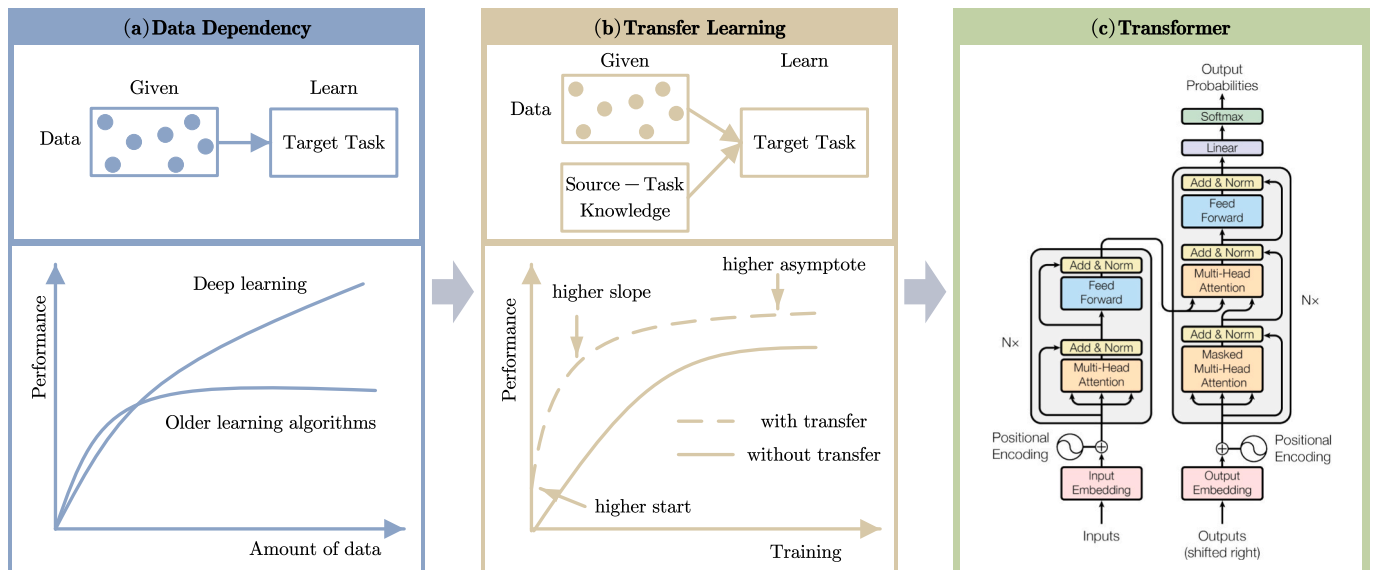


Fig. 20. Evolution from data dependency to transfer learning and transformer ((a)Data dependency; (b)Transfer learning [111]; (c)Transformer [112]).

remains a major challenge. These algorithms typically require large datasets for effective training, particularly in supervised learning models. The required data often include various sensor types, such as RGB images [107], point clouds [108], IMU [109], and depth images [23]. Obtaining construction datasets often faces challenges related to data confidentiality and commercial sensitivity. As a result, many researchers must create their own datasets, a process involving data collection, cleaning, annotation, verification, and evaluation. This process is not only time-consuming but also labor-intensive, significantly impacting research efficiency. While large public datasets are common in general fields, compiling similar datasets for deep learning in the construction domain is exceptionally difficult. Online image datasets contain far fewer images of built environments compared to everyday objects. Additionally, due to safety and security concerns, construction sites are typically closed to the public, making it harder to gather site-specific images.

Transfer learning offers a promising solution to the challenges of data dependency and distribution bias in deep learning, facilitating the broader application of computer vision in construction. This approach

involves leveraging a model trained on one task and adapting it to a different task. By transferring relevant knowledge from the pre-trained model and fine-tuning it with a small amount of domain-specific data, transfer learning helps address data scarcity while enhancing model performance in new tasks. In recent years, transfer learning has gained significant attention in machine learning and deep learning. Transfer learning is essential in construction, where several challenges arise in training deep learning models:

- **Big Data vs. Few Annotations:** While datasets are large, the number of labeled samples is limited, and data annotation is costly.
- **Big Data vs. Limited Computing Power:** Computing platforms often fall short of meeting the training demands of large datasets, necessitating the use of model migration.
- **Universal Models vs. Personalized Needs:** A single model cannot cater to the diverse requirements of different scenarios, making adaptation difficult.

To address these challenges, researchers can leverage transfer

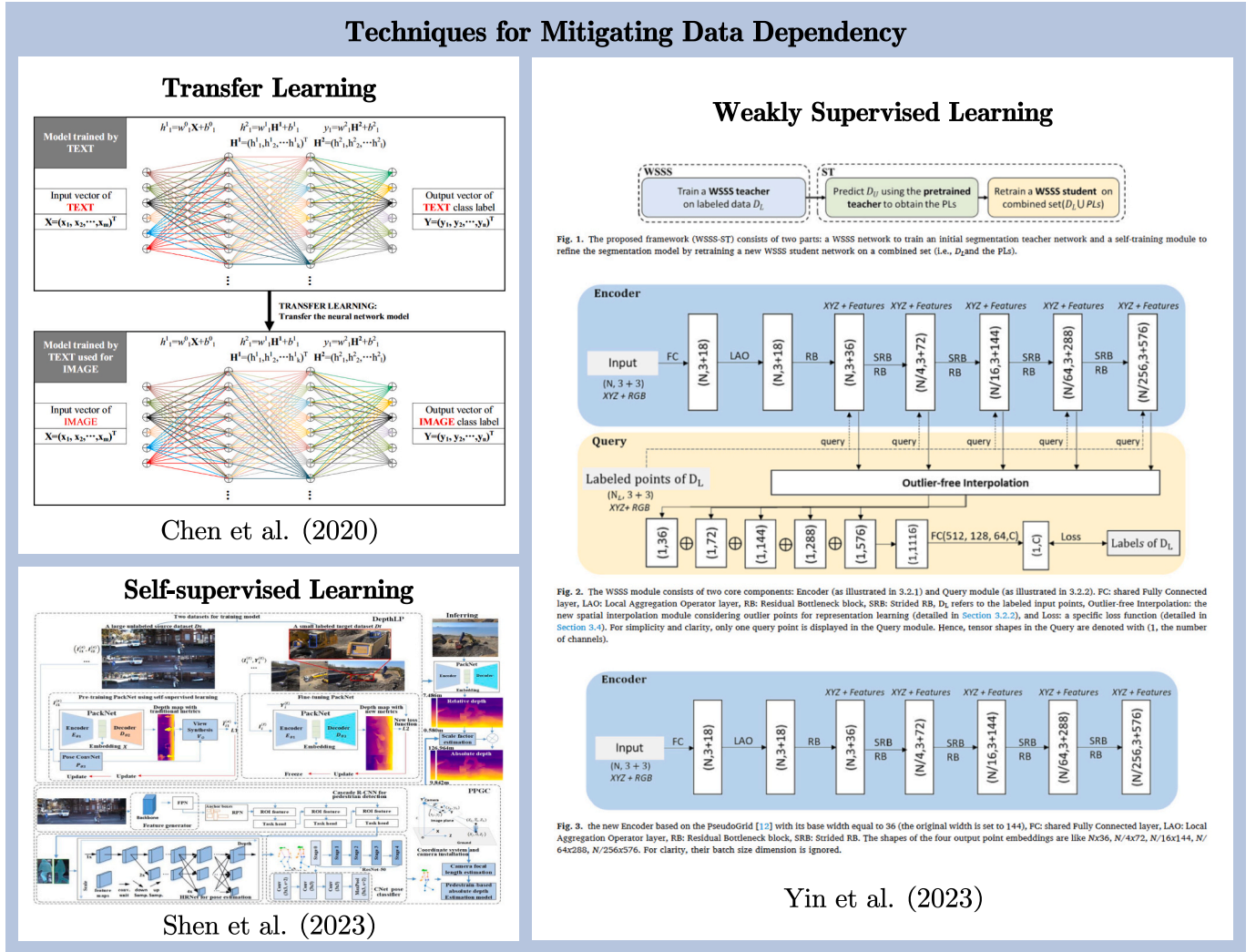


Fig. 21. Techniques for mitigating data dependency ((a)Transfer Learning [9]; (b)Self-supervised Learning [23]; (c)Weakly Supervised Learning [44]).

learning, which provides an effective approach to overcoming these contradictions. In 2020, Atkinson et al. [35] proposed a two-stage transfer learning method to tackle the challenge of capturing training images with specific entities like pipes and wires. In the first stage, they trained a model using the New York University (NYU) dataset, which includes a variety of general classes. This step allowed the model to learn general features from classes with more examples of pipes and wires. In the second stage, they fine-tuned the model using an underfloor scene dataset. Importantly, only the output layer was trained during this stage, while the region proposal network and backbone model remained unchanged. This ensured that the knowledge (such as network weights) gained in the first stage was effectively transferred and remained relevant for the second stage. Chen et al. [9] also applied a transfer learning approach to detect semantic regions in construction scenes. By using a pre-trained model as the starting point, they were able to reduce the amount of data needed for training and enhance the model's final performance. In 2023, Shen et al. [23] applied transfer learning to depth estimation models. For relative depth estimation, they pre-trained the DepthLP model on large-scale unlabeled datasets like KITTI and DDAD. The model was then fine-tuned for downstream tasks using a new loss function and only a small set of labeled images. In 2024, Luo et al. [110] proposed a novel interpretable transfer learning method for tunnel engineering risk prediction. This method identifies the source domain data that contributes most to transfer learning and quantifies its impact on improving prediction accuracy. By making the training process and the

role of source domain data transparent, interpretable transfer learning enhances the rational improvement of model performance.

Additionally, weakly supervised, self-supervised, and semi-supervised learning are effective techniques for addressing deep learning's dependency on large datasets. To reduce the need for manual annotations, active learning strategy can be employed to select a small subset of data for labeling. This approach leverages the model's uncertainty to identify the most informative samples. For example, in 2023, Yin et al. [44] introduced a novel weakly supervised semantic segmentation (WSSS) framework based on semantic query net (SQN), for large-scale industrial point clouds. This framework incorporates a self-training strategy to fully exploit the potential of unlabeled data. Similarly, Shen et al. [23] developed a new self-supervised learning paradigm for monocular depth estimation. They further transferred the pre-trained self-supervised model to other downstream tasks in construction scene understanding. Fig. 20 presents the development from data dependency to transfer learning and transformer, and Fig. 21 presents several techniques for mitigating data dependency.

4.5. From single-modal data and single technology to multi-modal fusion and multi-technology integration

(1) From single-modal data to multi-modal fusion

Although various types of data are collected from construction sites,

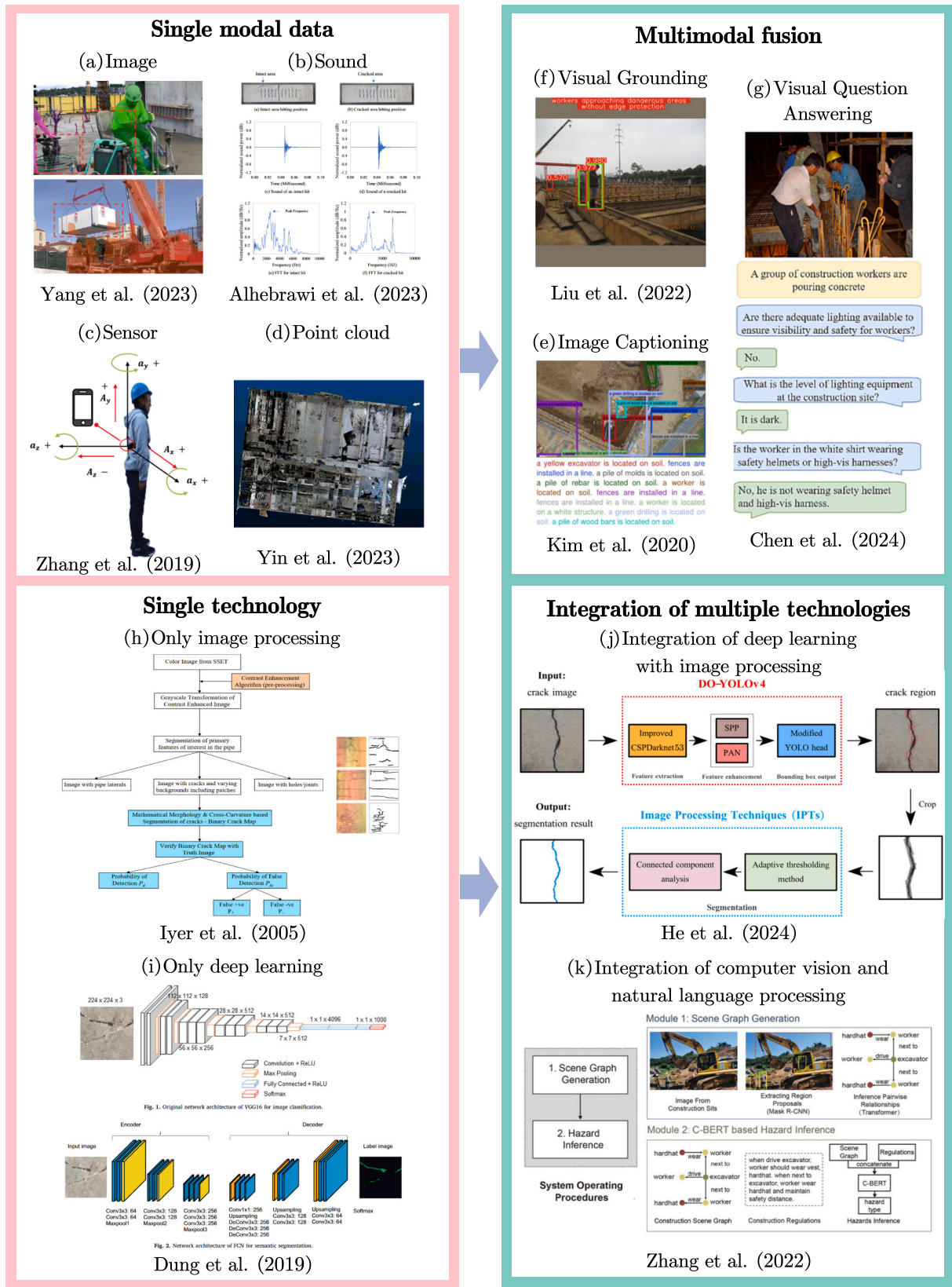


Fig. 22. Evolution from single-modal data to multi-modal fusion and from single technology to multi-technology integration ((a)Image [26]; (b)Sound [15]; (c) Sensor [75]; (d)Point cloud [44]; (e)Image Captioning [16]; (f)Visual Grounding [86]; (g)Visual Question Answering [14]; (h)Only image processing [113]; (i)Only deep learning [114]; (j)Integration of deep learning and image processing [106]; (k)Integration of computer vision and natural language processing [7]).

few studies have focused on multimodal data. The construction industry gathers and stores data in different formats, such as text, numerical data, images, videos, and audio, and applies machine learning to analyze data [21]. However, most studies tend to extract features from a single modality or data type. By combining multimodal features from different data sources, complementary information can be obtained, which can be integrated to enhance the performance of both shallow learning and deep learning. Current research often relies on two main types of devices for data collection, in addition to data from text or BIM. The first type includes sensors like accelerometers [75], gyroscopes [75], and IMUs [80], which capture physical or motion-related data. The second type involves various cameras, such as RGB, depth, stereo, and standard 2D cameras, which are typically installed on drones or fixed locations at construction sites. Image processing and computer vision are commonly used to analyze photos and videos. However, each sensor has its limitations. For instance, depth cameras have a restricted range, ground-based laser scanners struggle to map horizontal surfaces at the same level as the sensor, and ordinary cameras face challenges such as line-of-sight (LoS) issues, occlusion, and poor lighting conditions.

To overcome the limitations of a single sensor, combining different types of sensors can be a solution. However, this introduces added complexity in data processing and feature fusion. Compared to single-modal perception and reasoning, the use of multimodal perception and reasoning is still relatively limited. Image captioning [2,16,85,87–90], visual grounding [86], and visual question answering [14] are all typical multimodal tasks. Unlike a single visual feature map, natural language can convey high-level abstract concepts and richer semantic information. Since 2020, the fusion of vision and text has gained increasing attention from researchers, driving the shift from single-modal to multimodal. However, aside from vision and text fusion, there are relatively few studies exploring other modalities. In the future, the feasibility and potential applications of combining vision with other modalities should be explored, such as RGB images with depth images [36,38], images with BIM [43], images with point clouds, or vision with hearing [19]. Therefore, conducting research based on multimodal data is crucial to enhancing construction scene understanding. Future studies should focus on extracting multimodal features and developing effective feature fusion mechanisms to improve the performance of deep learning in this field.

(2) From single technology to multi-technology integration

The technologies used in construction scene understanding include, but are not limited to, image processing, machine learning, deep learning, building information modeling, ontology, and natural language processing. Early studies mostly relied on a single technology to achieve scene understanding. However, each technology has its limitations, and using only one cannot provide a comprehensive understanding of the scene. With advancements in computer technology and hardware, the integration of multiple technologies has increasingly gained the attention of researchers.

Traditional image processing typically relies on manually designed algorithms to extract features and process images, whereas deep learning automatically learns features and patterns by training neural networks. The main advantage of deep learning is its ability to learn features and patterns without manual design. Additionally, deep learning models often achieve higher accuracy and performance, making them suitable for more complex tasks. However, deep learning also faces several challenges, including poor interpretability, limited controllability, high computational resource demands, limited generalization ability, and data privacy concerns. In 2021, Arashpour et al. [1] suggested that combining image processing with deep learning could optimize performance indicators such as time, cost, and quality in construction scene understanding tasks. In 2022, Fang et al. [71] proposed a method for retrieving images of unsafe worker behaviors. This study used a deep learning feature extraction method and three image

processing feature extraction methods, namely color-based, texture-based, and shape-based methods, and fused the four extracted features according to certain weights. In 2024, He et al. [106] proposed a hybrid crack segmentation method that combines the strengths of CNN-based object detection and image processing-based segmentation. First, DO-YOLOv4 is used to locate the crack area, and then image processing techniques detect the crack pixels within that area. This approach reduces labeling costs while improving both the efficiency and accuracy of crack segmentation. Additionally, the integration of computer vision with technologies such as BIM, ontology, and natural language processing has gained increasing attention from researchers. In 2019, Xiong et al. [24] integrated visual relationship detection with ontology to automatically identify hazards on construction sites. In 2020, Zhang et al. [62] combined object detection and ontology to identify risks in construction. In 2022, Zhang et al. [7] proposed an automatic hazard inference method that combines relationship detection and natural language processing, using scene graphs and C-BERT. In 2024, Chen et al. [14] proposed the Visual Construction Safety Query (VCSQ) system. This system integrates real-time image captioning (IC), safety-centric visual question answering (VQA), and keyword-based image-text retrieval (ITR) with head-mounted augmented reality (AR) devices.

Data from different modalities represent various types of information in real-world scenes and require different processing technologies. Each technology also has its own strengths and limitations. For example, deep learning for image data can automatically learn complex features but requires significant computing resources and time. In contrast, traditional image processing can reduce data labeling costs and lower computing power requirements. To enhance the intelligence of construction scene understanding, scholars need to explore how to effectively integrate the advantages of these different technologies. Fig. 22 presents the development from single-modal data and single technology to multi-modal fusion and multi-technology integration.

5. Challenges and future directions

5.1. Situation awareness (SA)

Situation Awareness is the ability to perceive elements in the environment, understand their meaning, and predict their state in the near future [115]. Construction sites are full of hidden dangers, and workers' inability to accurately perceive risks can lead to missed or misidentified hazards, increasing the likelihood of accidents. For example, if workers fail to recognize changes in road conditions, they may not adjust their movements accordingly, further raising the risk of accidents [116]. This ability to detect, perceive, and understand a changing environment in real-time is called "situation awareness" [117]. An important factor affecting situation awareness is the limited capacity of the attention system [117]. To maintain situation awareness in hazardous environments, construction workers must visually scan for potential hazards, gather feedforward information about risks, and detect impending dangers [118]. However, in a dynamic construction environment, human attention is limited. Excessive demands on attention can prevent workers from fully maintaining situation awareness, increasing the risk of construction accidents [117]. In 2016, Hasanzadeh et al. [119] pioneered the use of mobile eye-tracking technology to measure the situation awareness levels of construction workers. In 2018, Hasanzadeh et al. [118] used mobile eye-tracking technology to investigate the relationship between construction workers' visual attention allocation and their level of situation awareness in the presence of fall and trip hazards. In 2020, a study by Choi et al. [120] found that forklift operators' situation awareness of other workers was significantly influenced by the complexity of the subtask they were performing at the time. In 2022, Han et al. [121] studied the factors influencing the situation awareness levels of highway workers, providing a foundation for designing human-centered hazard warning methods tailored to different work environments.

In the era of big data, construction sites generate vast amounts of information, and the fast-changing conditions pose significant challenges to situation awareness. However, advancements in computer science have greatly enhanced computers' ability to process large datasets and reason within set rules. This improvement helps shorten the observe-orient-decide-act (OODA) cycle, enabling humans to better perceive their environment, make decisions, and mitigate risks. In 2020, Zhu [122] combined convolutional neural networks, situation awareness, and database technologies to study road traffic situation awareness systems. The research focused on the processes of information collection, processing, and analysis within such systems. However, there is limited research on the application of situation awareness in construction using digital technologies. In 2008, Katz et al. [123] suggested using a camera network to track on-site equipment and materials, helping to enhance situation awareness in construction scenes. Later, in 2018, Fang et al. [124] introduced a new operator assistance system designed to improve mobile crane operators' situation awareness of safety risks. In 2023, Zhang et al. [125] reviewed the role of situation awareness and digital technology in construction safety management, offering a theoretical foundation for developing a situation awareness system for construction sites. Based on the above research, it is clear that situation awareness in construction scenes is still at a theoretical stage, with significant potential for further exploration. In the future, situation awareness in construction scenes, driven by digital technology and deep learning, is expected to be a key research trend.

5.2. Three-dimensional (3D) scene understanding

Previous studies have highlighted the limitations of using 2D geometric distances for relationship analysis [4]. As a result, 3D object detection [6], 3D pose estimation [82], and 3D semantic segmentation [38,44,45] have emerged as key research trends. Future research on scene understanding will focus on collecting 3D data from construction scenes, creating training datasets for 3D deep learning, and developing algorithms that process 3D data and spatial relationships to improve the robustness and accuracy of construction scene understanding. In 2020, Yan et al. [6] proposed a method to reconstruct 3D candidate boxes for detected objects, such as workers and heavy vehicles, using monocular 2D camera videos. In 2022, Mascaro et al. [38] combined RGB images with depth information to develop a 3D instance segmentation algorithm. In 2023, Yin et al. [44] introduced a large-scale industrial point cloud segmentation algorithm based on weakly supervised learning. You et al. [45] proposed an enhanced object segmentation method that uses a low-resolution LiDAR scanner for high-precision object boundary recognition. Liu et al. [82] applied 3D posture estimation to detect workers at risk of falling.

Thanks to the development of long-range sensors like LiDAR and short-range sensors such as structured light, acquiring 3D point cloud data has become increasingly convenient. Compared to 2D images, point cloud data is less affected by external lighting and distance, and more accurately reflects the spatial structure of the real world. It provides richer geometric, shape, and scale information. With these advantages, 3D scene understanding and reconstruction technologies allow computers to record and comprehend the 3D world, which is crucial for industrial production automation and management informatization. These technologies have broad applications in areas like scene reconstruction, SLAM, and robot perception. As a result, numerous studies have focused on key issues such as point cloud feature extraction, matching fusion, semantic segmentation, and point cloud completion, achieving significant progress.

However, there are still challenges, such as significant gaps in scanning scenes and high computational costs for high-precision 3D scenes, which impact the accuracy of applications. First, point cloud data is unstructured, and real objects come in various shapes, meaning the completion method must be robust and generalizable. Second, for workers in 3D scenes, it's essential to explore the connection between

the scene and the workers' behavior. To advance related technologies and facilitate practical applications, further research is needed in areas like segmentation of construction scene point clouds with limited annotations, shape and texture completion for large-scale construction scenes, and understanding and modeling worker behavior in 3D environments.

5.3. Enhancing dynamic scene understanding using time series analysis

Construction scenes are dynamic, with constant changes in people, materials, machinery, and the working environment. These dynamic characteristics present technical challenges for computer vision, such as dynamic object segmentation, occlusion, and camera positioning. While current research can handle static objects, more effective strategies are needed to manage dynamic objects. For example, ORB-SLAM [126] may lose track when moving objects are present. Additionally, an action or event often involves a sequence of actions with temporal relationships. The use of time series analysis offers a more accurate way to describe construction events. For instance, the actions of "carrying" and "handling" may appear similar in terms of spatial and visual characteristics in a static image. However, to distinguish whether a worker is "carrying" or "handling" an object, time series analysis—such as tracking spatial distance and Interaction over Union (IoU) over time—can be useful [17].

In 2019, Cai et al. [8] used long short-term memory networks (LSTM) to analyze work groups in construction activities. Given the dynamic nature of construction sites, the grouping results change over time, making this study a more accurate representation of real-world construction scenes. In 2023, Chen et al. [25] applied time series analysis to reduce misidentifications caused by environmental factors or occlusion, enhancing hazard identification performance. In 2024, Dzung et al. [83] used a recurrent neural network to classify IMU time series data, reducing false alarms for events like tripping, falling, and coma. To improve the accuracy of dynamic object segmentation, Yang et al. [26] applied bi-directional propagation to the fused mask, enhancing the consistency of the dynamic object mask in the temporal context. Future research on construction scene understanding could incorporate time series analysis models to extract deeper semantic information, offering more comprehensive support for downstream tasks like safety management and hazard prediction.

5.4. Data generation for construction scene understanding

Current datasets for construction scene understanding are usually limited to specific stages or scenarios, leading to several issues:

- Models trained on static images from a single stage may struggle to understand the dynamic nature of construction sites, where elements change as work progresses.
- The complexity of construction sites and the limited coverage of cameras can result in many objects being obstructed, making it difficult to label or detect them.
- Site conditions (such as inaccessibility to dangerous areas) or regulations (such as drone restrictions) can make it challenging to capture images from specific camera angles.
- Due to practical limitations in data collection, relying solely on real-world images may not capture the full range of construction object characteristics (such as varying postures, textures, and colors).
- Real images may also raise concerns about personal privacy and sensitive information.

In this context, image synthesis methods can be used to generate data for construction scene understanding. Virtual reality technologies, such as Building Information Modeling (BIM), can simulate real-world scenes, allowing virtual cameras to capture images from any perspective and synthesize training datasets for specific needs. Additionally, data

generation techniques like generative adversarial networks (GANs) can help address issues arising from insufficient datasets. In 2021, Hong et al. [127] proposed a high-quality synthetic data generation method with automatic annotation for infrastructure scenes. First, CycleGAN was used to convert BIM images into real-world images. Then, the spatial information in BIM was used to automatically annotate the images, creating various synthetic datasets. Finally, the selected synthetic datasets were combined to form the final synthetic dataset. In 2022, Kim et al. [128] explored the potential of using synthetic data for developing visual models. They trained a deep neural network (DNN) using only synthetic data and evaluated its performance on a worker detection task. Wei et al. [129] used BIM semantics and progress information, along with the Unreal Engine, to generate image datasets for construction under varying environmental conditions (seasons, lighting, weather) and different construction stages. In 2023, Kim et al. [130] proposed a construction worker-centric image synthesis method that automatically generates and labels synthetic images of workers in various poses, performing different activities, and under different outdoor conditions. Tang et al. [131] proposed a method to automatically generate synthetic point clouds from BIM and evaluated their potential for improving indoor scene understanding. Ying et al. [132] proposed a method for generating synthetic images based on BIM. This method captures realistic images of various scenes from BIM and automatically generates annotations for 12 building scene understanding (BSU) tasks, including pose estimation, scene recognition, depth estimation, object detection, semantic segmentation, instance segmentation, and panoptic segmentation.

Currently, construction datasets are mainly used for academic purposes. In contrast, synthetic image datasets can help avoid issues related to data confidentiality and licensing, while also promoting the application of scene understanding technology in construction. Future research can further explore the potential of synthetic images in construction, addressing challenges posed by the dynamic and complex nature of construction sites.

5.5. Ethical issues of data collection and privacy computing

Data privacy is a critical concern in the age of the Internet of Things. Researchers and practitioners collect various types of data, such as materials, equipment and worker locations, and worker health status, using different sensors. While some advocate for automated workplace monitoring to serve organizational interests, others argue that it can increase worker stress by invading their privacy. It is essential that individuals are informed about the existence of monitoring, the reasons behind data collection, and their rights in managing their personal information. Moreover, clear policies and procedures should be in place to address potential privacy issues, including obtaining informed consent from workers, anonymizing data whenever possible, and ensuring the secure storage and processing of collected information.

Privacy computing refers to technologies that enable multiple parties to jointly analyze and compute data without compromising its confidentiality. It involves the description, measurement, evaluation, and integration of privacy-related information when processing various types of data, including video, audio, images, graphics, and text. This process forms a set of formalized privacy computing theories, algorithms, and applications, with quantitative evaluation standards, to support the protection of privacy in multi-system environments. As a key approach to balancing data sharing and privacy protection, privacy computing can effectively overcome “data island” barriers, allowing for data openness and sharing while ensuring security. There are three main technologies driving privacy computing:

- Cryptography-based methods, represented by secure multi-party computation (MPC);
- The integration of Artificial intelligence and privacy protection, represented by federated learning (FL);

- Trusted hardware-based technologies, represented by trusted execution environments (TEE).

Secure multi-party computation (MPC) enables multiple participants in a network, without the need for a trusted third party, to collaborate on computing a function and obtain verifiable results. Each participant holds private input data, and the computation process uses interactive, irreversible encryption to ensure that participants can only access the output related to their own data. This method prevents any participant from inferring the private input data or output results of others. Federated Learning (FL) is an emerging machine learning approach that enables training across multiple distributed clients with local datasets, while aggregating the results on a centralized server. By allowing machine learning tasks to be performed collaboratively without transmitting the original data, federated learning offers a promising solution to the conflict between user privacy and data sharing. A Trusted Execution Environment (TEE) is a security architecture that combines both software and hardware to create a secure computing environment isolated from the outside world. It achieves this by utilizing the central processing unit (CPU) or allocating a portion of memory as a secure area. Sensitive data is processed within this secure area, and without authorized access, no external attacker—including system administrators—can control the operations or access the sensitive data. The hardware isolation ensures the complete security of privacy-sensitive computations within the environment.

In construction, in 2020, Cheng et al. [133] proposed an encryption key distribution strategy for construction projects. This strategy enables project participants to authenticate their identities and securely share sensitive information between parties. In 2021, Kochovski et al. [134] integrated blockchain into smart construction applications to provide trusted access to artificial intelligence methods, while also supporting privacy protection regulations. Li et al. [135] used federated learning to aggregate the gradient parameters of image data from each smart work packaging (SWP) and developed a more personalized model through transfer learning. In this approach, each SWP trains its local model on a local dataset, without sharing the data itself. Instead, only the updated parameters of the local model are submitted to the cloud. In 2022, Xu et al. [136] proposed a blockchain-based framework for occupational health and safety (OSH) management that protects workers' privacy. This framework minimizes the risk of privacy violations while ensuring that OSH management is not disrupted. In 2023, MOE et al. [137] proposed a worker safety prediction method using federated learning. This approach aims to reduce the risk of accidents and fatalities while ensuring the security of sensitive data. Saah et al. [138] used Hyperledger Fabric blockchain to store personal and security data of construction workers. Their work not only showcased blockchain's potential in addressing construction data security challenges but also highlighted its practical applicability through extensive testing. In 2024, Cai et al. [49] applied federated learning to a human intention prediction model for human-computer interaction. The study demonstrated that, while protecting data privacy, federated learning achieves prediction accuracy comparable to centralized training and significantly outperforms local training. Gugssa et al. [139] used edge computing to enhance the real-time performance of PPE detection while protecting the data privacy of construction workers. Their results showed that edge computing not only reduces latency but also ensures data security by eliminating the need to transmit all data to the cloud server.

Data security is a crucial factor in determining whether a technology can be successfully implemented in engineering practice. However, in the field of intelligent construction, there is limited focus on data privacy and security. Future research should explore ways to better protect data privacy and security, thereby improving the efficiency of information management in construction.

5.6. Interpretability level of deep learning frameworks

Deep learning is a subset of machine learning, distinguished by its ability to mimic the structure and mechanism of the human brain's neural network. Through multi-level neuron interconnections and information transmission, it can learn, represent, and predict complex data patterns. However, the highly nonlinear and high-dimensional nature of deep learning models often turns their prediction process into a "black box," making it difficult to understand the underlying decision-making logic. In computer science, the term "black box" refers to a data-driven algorithm that generates useful information without revealing its internal workings [140]. Data-driven machine learning and deep learning models excel at identifying patterns and associations in data, but they cannot guarantee causality. Relying on a system that cannot explain its decisions is risky for decision-makers [141]. Therefore, the inherent complexity and lack of transparency in deep learning models limit their credibility and usability, especially in critical applications where decision-making must be explained and verified, such as medical diagnosis, financial risk assessment, and industrial system monitoring. As a result, researchers have begun exploring ways to improve the interpretability of these models, enabling a better understanding of their logical rules, decision-making processes, and internal mechanisms, ultimately enhancing trust in the models.

Explainable Artificial Intelligence (XAI) can be classified into two methods: ante-hoc interpretability and post-hoc interpretability [142,143]. Ante-hoc interpretability refers to built-in methods that provide explanations before model training, such as decision trees, linear regression, and rule-based models. In contrast, post-hoc interpretability involves analyzing the model's key features after training to understand its behavior and decision-making processes.

In the field of civil engineering, in 2023, Gao et al. [144] proposed an explainable spatiotemporal graph convolutional network for clogging detection. This network is used to assess the risk level of clogging for each ring. By incorporating complex network indices, the model provides more detailed explanations, demonstrating the potential value of learning from explainable AI. Wang et al. [145] developed the causal explainable gated recurrent unit (CX-GRU) for real-time prediction of TBM parameters, addressing the black box problem in AI. The study found that the causal discovery component improved the computational efficiency of model training by an average of 8.8 %. In 2024, Luo et al. [110] proposed a novel interpretable transfer learning method for risk prediction in tunnel construction. This method identifies the source domain data that contributes most to transfer learning and quantifies its impact on improving prediction accuracy.

Although explainable AI is gaining attention in fields such as law and medicine, it has received relatively limited focus in the construction industry. However, the introduction of explainable AI could open new research directions and help address practitioners' concerns and hesitations about AI applications [146]. In future research, AI development is likely to shift towards decision-making, and the focus should move from model accuracy to model interpretability.

6. Conclusions

Over the past two decades, scene understanding has become a growing research focus in construction, achieving significant progress. However, there is still a lack of a systematic definition of the concept in academia. To address this gap, this paper provided a comprehensive review of construction scene understanding, with a focus on construction workers. It analyzed development trends and highlighted the gaps and limitations in current research.

Through the content-based review, the paper outlined the development of construction scene understanding into two key aspects: (1) perception ability and (2) reasoning ability. This outline is based on five scene elements—signals, pixels, objects, relationships, and events—and the richness of semantic information, ranging from low-level to high-

level semantics. Additionally, the research background, technical approach, and progress in each sub-field of these two aspects are discussed, highlighting the significant potential of scene understanding applications in construction.

This paper provided a comprehensive review of relevant literature, analyzing five key research trends in construction scene understanding. First, with the advancement of artificial intelligence and large language models, algorithms have evolved from being tailored for specific engineering tasks and application scenarios to more general AI models applicable to construction. Second, scene understanding has progressed from perceiving entities to perceiving relationships, from single object to complex objects, and from single relationship to complex relationships. Third, scene understanding algorithms have evolved from rule-based learning to end-to-end learning. Fourth, to address the data dependency challenges in deep learning, researchers are increasingly focusing on techniques like transfer learning and self-supervised learning. Finally, the fusion of multi-modal data and the integration of multi-technology is expected to enhance key performance indicators, such as time, cost, and quality, in construction scene understanding tasks.

Although the research on construction scene understanding has made significant progress, many challenges remain. Based on the review of relevant literature, this paper proposed six future development directions, aiming to inspire further research. These include (1) Situation awareness of construction scenes; (2) Three-dimensional scene understanding; (3) Time series analysis of dynamic scenes; (4) Data generation for construction scene understanding; (5) Privacy computing; (6) Interpretability level of deep learning frameworks and explainable artificial intelligence.

In summary, construction scene understanding needs overall improvement to effectively address the dynamic and ever-changing nature of construction sites and the complexity of downstream tasks. Given the rapid advancement of related technologies in artificial intelligence, it is expected that these challenges will be overcome in future research, ultimately enhancing the intelligence of construction scene understanding and improving the automated management of construction safety, progress, quality, productivity, and resources.

CRedit authorship contribution statement

Huimin Li: Writing – original draft. **Hui Deng:** Writing – review & editing. **Yichuan Deng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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